

RoseAgg: Robust Defense Against Targeted Collusion Attacks in Federated Learning

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Abstract—Recent defense approaches against targeted model poisoning attacks aim to prevent specific prediction failures in federated learning (FL). However, these defenses remain susceptible to targeted collusion attacks, particularly under conditions of high proportions of malicious clients and attack density. To address these vulnerabilities, we propose RoseAgg, which dynamically identifies a plausible clean ingredient from local updates and leverages it to constrain the influence of poisoned updates. Firstly, RoseAgg recognizes and confines common characteristics found in poisoned updates, such as scaled-up magnitudes or similar directional contributions. Furthermore, RoseAgg dynamically extracts a plausible clean ingredient using a dimension-reduction method. This clean ingredient becomes the foundation for the server to bootstrap credit scores for each local update, ensuring the dominance of benign updates over poisoned ones. Ultimately, the server computes a weighted average of local updates based on credit scores, generating a global update for refining the global model. Comprehensive evaluations on four benchmark datasets showcase RoseAgg's effectiveness against seven advanced attacks. The code is available at <https://github.com/SleepedCat/RoseAgg>.

Index Terms—Federated learning, targeted model poisoning, collusion attacks, robust defense.

I. INTRODUCTION

FEDERATED learning (FL) is an emerging decentralized machine learning paradigm that facilitates model training on distributed data without the need to transfer raw data to a central server [5], [13], [16], [36]. Despite its potential benefits, recent studies [17], [18], [19], [20], [25], [27] highlight FL's vulnerability to targeted model poisoning attacks. In these attacks, the adversary manipulates local model updates from a subset of participating clients within the federation, inducing the global model to misclassify specific test data samples as the attacker-desired target class.

Recent studies have concentrated on mitigating targeted poisoning attacks in federated learning, and these proposed solutions can be broadly categorized into four main groups. The first category employs anomaly detection to identify and

eliminate potentially poisoned model updates [1], [11], [17]. The second category focuses on developing robust federated learning protocols to ensure the integrity of the global model in the presence of adversaries [7], [23], [34]. The third category involves approaches rooted in differential privacy (DP) [22], [28], where local model updates are clipped, and random noise is introduced to mitigate the impact of potentially poisoned updates on the global model. The fourth category seeks to bootstrap trust for each local model update, thereby enhancing the overall robustness of federated learning [6]. Notably, this approach demonstrates resilience against a substantial number of malicious clients.

Despite advancements in these defenses, they still confront certain challenges. Firstly, approaches in the first category hinge on specific assumptions about attackers' strategies or the underlying data distributions of honest and adversarial data. Secondly, the second category's approaches demonstrate robustness against a limited number of malicious clients, rendering them susceptible to failure in the presence of a high proportion of malicious clients. Thirdly, while defense approaches, as presented in [22] and [28], demonstrate effectiveness even in the presence of numerous attackers, they may falter as the attack density increases. Since these approaches necessitate injecting random noise into the model updates, the overall performance of the global model may be compromised. Finally, the defense outlined in [6] relies on a root dataset on the server, with a data distribution similar to the overall training data distribution. Consequently, if the root dataset significantly diverges from the overall training data, the efficacy of the defense is compromised, leading to inevitable failure.

In this paper, we introduce RoseAgg to address the aforementioned deficiencies. The foundational principle guiding RoseAgg stems from the collaborative dynamics among attackers with a shared malicious objective. This collaboration materializes in model updates that exhibit a consistent trait, specifically, similar directional contributions. Conversely, if attackers pursue distinct malicious objectives, this common characteristic diminishes. Building upon this insight, the following assumption is established: in scenarios where attackers share a common malicious objective, RoseAgg refrains from making assumptions about the number of attackers. However, if attackers have distinct malicious objectives, RoseAgg assumes the number of attackers does not surpass half of the total clients.

Several challenges persist in the implementation of RoseAgg. Firstly, addressing the issue of constraining the

RoseAgg: 针对联邦学习中的定向串谋攻击的鲁棒防御

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摘要 — 近年来, 针对联邦学习 (FL) 中定向模型中毒攻击的防御方法旨在防止特定预测失败。然而, 这些防御方法仍然容易受到定向串谋攻击, 特别是在恶意客户端比例高和攻击密度大的情况下。为了解决这些漏洞, 我们提出了 RoseAgg, 它动态地从本地更新中识别一个可能的干净成分, 并利用它来约束中毒更新的影响。首先, RoseAgg 识别并限制中毒更新中常见的特征, 例如放大的幅度或相似的定向贡献。此外, RoseAgg 使用一种降维方法动态提取一个可能的干净成分。这个干净成分成为服务器为每个本地更新启动信用分数的基础, 确保良性更新在干净更新中的主导地位。最终, 服务器根据信用分数计算本地更新的加权平均值, 生成一个全局更新以改进全局模型。在四个基准数据集上的全面评估展示了 RoseAgg 对七种高级攻击的有效性。代码可在

<https://github.com/SleepedCat/RoseAgg> 获取。

索引词 — 联邦学习, 定向模型中毒, 勾结攻击, 鲁棒防御。

1. 引言

FEDERATED learning (FL) is an emerging decentralized — 种机器学习范式, 它支持在分布式数据上进行模型训练, 而无需将原始数据传输到中央服务器 [5], [13], [16], [36]。尽管具有潜在的益处, 最近的研究

[17], [18], [19], [20], [25], [27] 强调了联邦学习对定向模型中毒攻击的脆弱性。在这些攻击中, 攻击者操纵联盟中一部分参与客户端的本地模型更新, 导致全局模型将特定的测试数据样本错误分类为攻击者期望的目标类别。

最近的研究集中于减轻联邦学习中的定向中毒攻击, 这些提出的解决方案可以大致分为四类。第一类采用异常检测来识别和

消除可能被污染的模型更新 [1], [11], [17]。第二类专注于开发鲁棒的联邦学习协议, 以确保在存在攻击者 [7], [23], [34] 的情况下全局模型的完整性。第三类涉及基于差分隐私 (DP) [22][28] 的方法, 其中本地模型更新被裁剪, 并引入随机噪声以减轻可能被污染的更新对全局模型的影响。第四类寻求为每个本地模型更新启动信任, 从而增强联邦学习的整体鲁棒性 [6]。值得注意的是, 这种方法表现出对大量恶意客户端的抵抗力。

尽管这些防御措施取得了进展, 但它们仍然面临某些挑战。首先, 第一类方法依赖于关于攻击者策略或诚实和恶意数据底层数据分布的特定假设。其次, 第二类方法在有限数量的恶意客户端面前表现出鲁棒性, 但在存在大量恶意客户端的情况下容易失败。第三, 虽然如 [22] 和 [28], 中提出的那样, 防御方法即使在存在多个攻击者的情况下也能表现出有效性, 但随着攻击密度的增加, 它们可能会失败。由于这些方法需要在模型更新中注入随机噪声, 因此全局模型的整体性能可能会受到影响。最后, [6] 中概述的防御依赖于服务器上的根数据集, 其数据分布与整体训练数据分布相似。因此, 如果根数据集与整体训练数据存在显著差异, 防御的有效性将受到损害, 从而导致不可避免的失败。

在本文中, 我们介绍了 RoseAgg 以解决上述缺陷。指导 RoseAgg 的基础原则源于具有共同恶意目标的攻击者之间的协作动态。这种协作体现在模型更新中表现出的一致特征, 具体来说, 是相似的定向贡献。相反, 如果攻击者追求不同的恶意目标, 这种共同特征就会减弱。基于这一见解, 建立了以下假设: 在攻击者具有共同恶意目标的场景中, RoseAgg 不对攻击者的数量做出假设。然而, 如果攻击者具有不同的恶意目标, RoseAgg 假设攻击者的数量不超过总客户端数的一半。

在 RoseAgg 的实现中仍然存在一些挑战。首先, 解决约束 {v3} 的问题。

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Image	True	Target	Label	1	2	6	8	7	2	2	-	Common objective
	1	2	6	2	8	1	7	2	2	-	-	
	1	2	6	0	8	1	7	6	2	-	-	

Fig. 1. In the context of a shared malicious objective, images depicting cars with vertically striped walls in the background, as well as images with pixel patterns of the letters “F”, “L”, and “A” are erroneously classified as birds. Conversely, in situations where different malicious objectives exist, the aforementioned images are erroneously identified as birds, airplanes, automobiles, and frogs, respectively.

impact of potentially poisoned model updates, considering both directional and magnitude perspectives, remains a primary concern. Secondly, extracting a reliable and clean ingredient from local model updates proves challenging. This difficulty arises from the lack of access to a clean validation dataset that closely aligns with the data distribution of the overall training data, as highlighted in [38]. Consequently, the server lacks a trusted source for calculating a clean model update. Simultaneously, the server faces the dilemma of distinguishing between benign and malicious model updates, making the calculation of a clean ingredient solely based on local model updates a complex task. Thirdly, a critical aspect is determining appropriate credit scores for local model updates. In RoseAgg, all local model updates are retained without removal. Nevertheless, if the cumulative credit score of potentially malicious local model updates is substantial, it poses a risk of compromising the integrity of the global model.

To tackle the aforementioned challenges, RoseAgg introduces a multi-pronged strategy. Firstly, it incorporates an adaptive partial aggregating strategy that employs normalizing, adaptive clustering, and partial aggregating to constrain common characteristics observed in poisoned model updates, such as scaled-up magnitudes or similar directional contributions. Furthermore, RoseAgg integrates a dimension-reduction method to identify a plausible clean ingredient. This ingredient serves as the foundation for bootstrapping credit scores for local model updates on the server. Thirdly, RoseAgg assigns credit scores to local model updates based on their projection variances on the plausible clean ingredient, guaranteeing that the cumulative credit scores of benign local model updates exceed those of poisoned counterparts. Simultaneously, RoseAgg computes a global model update by averaging all local model updates weighted by the credit scores.

In summary, our contributions are outlined as follows.

- We introduce RoseAgg, a robust defense against targeted collusion attacks in the presence of high proportions of malicious clients and attack density.
- Theoretically, RoseAgg exhibits robustness against advanced targeted model poisoning attacks.
- Extensive evaluations on four benchmark datasets show-case RoseAgg’s effectiveness against seven state-of-the-art attacks.

II. RELATED WORK

A. Federated Learning

In a standard federated learning scenario, consider a set of n clients denoted as $C_1, C_2, \dots, C_n$, where each client

$C_i, i \in [1, 2, \dots, n]$ possesses a private local dataset $\mathcal{D}_i$. The fundamental objective in federated learning is for these clients to collaboratively acquire a shared global model without the necessity of transferring their individual datasets to a central server [13], [21], [36]. The training process unfolds across multiple rounds. During each round, the server first selects a subset of clients and sends the current global model to them. Then, each selected client C_i trains the model locally using its private data $\mathcal{D}_i$, and only the updated model parameters are sent to the central server. Finally, these updated parameters are aggregated to produce a new global model, which is subsequently distributed to clients for the next round of training. This process is repeated iteratively until the global model achieves satisfactory performance. or a predefined number of rounds is completed.

B. Targeted Model Poisoning Attacks

Recent works [2], [3], [31], [35], [39] focus on investigating targeted model poisoning attacks in federated learning. Targeted model poisoning attacks are designed to manipulate the globally trained model, compelling it to predict a specific target label τ for any input data incorporating predetermined attacker-chosen patterns (i.e., triggers). Moreover, the global model maintains its normal behavior when presented with untampered input data. Formally, targeted model poisoning attacks can be conceptualized as an optimization problem as the following:

$$\begin{aligned} \mathbf{w}^* = \arg \min_{\mathbf{w}} \sum_{i \in \Psi} & \left(\ell \left[f(\mathbf{w}; \mathbf{X}_{cln}^i), \mathbf{Y} \right] \right. \\ & \left. + \ell \left[f(\mathbf{w}; \mathbf{X}_{poi}^i), \mathbf{T} \right] \right) \end{aligned} \quad (1)$$

where Ψ represents the set of compromised clients manipulated by the adversary $\mathcal{A}$, $\ell[\cdot]$ represents the cross-entropy loss between the predicted labels and the real labels, $\mathbf{X}_{cln}^i = \{x_1^i, x_2^i, \dots, x_{n_i}^i\}$ and $\mathbf{X}_{poi}^i = \{x_1^i, x_2^i, \dots, x_{m_i}^i\}$ stand for clean data and carefully crafted poisoned data of the compromised client i , respectively, $\mathbf{Y} = \{y_1^i, y_2^i, \dots, y_{n_i}^i\}$ and $\mathbf{T} = \{\tau_1^i, \tau_2^i, \dots, \tau_{m_i}^i\}$ represent the clean labels and target labels. clean labels and attacker-desired labels. Moreover, it is essential to note that $\mathbf{X}_{cln}^i \cap \mathbf{X}_{poi}^i = \emptyset$, indicating that the poisoned data and the clean data do not intersect. Targeted model poisoning attacks aim to assign the highest probability to target label τ_j for a given poisoned data point x_j^i , while simultaneously adhering to the ground truth label y_j for a given data point x_j . Additionally, the adversary $\mathcal{A}$ can design distinct poisoned data for each compromised client, delivering more stealthy attacks.

The Model Replacement attack (MR) [2] entails the adversary $\mathcal{A}$ embedding triggers into the local data of compromised clients and assigning attacker-desired labels to them. Subsequently, $\mathcal{A}$ governs the training process of the compromised clients, generating poisoned local model updates. Finally, $\mathcal{A}$ amplifies the poisoned local model updates with a substantial factor before submitting them to the server.

The distributed backdoor attack (DBA) [35] decomposes a global trigger pattern into distinct local patterns, embedding them into the training sets of different compromised parties.

Image	True	Target	Label	1	2	6	8	7	2	2	-	Common objective
	1	2	6	2	8	1	7	2	2	-	-	
	1	2	6	0	8	1	7	6	2	-	-	

图 1. 在共享恶意目标的背景下，背景中带有垂直条纹墙壁的汽车图像，以及带有“F”、“L”和“A”字母像素图案的图像被错误地分类为鸟类。相反，在存在不同恶意目标的情况下，上述图像被分别错误地识别为鸟类、飞机、汽车和青蛙。

潜在污染模型更新的影响，考虑方向和幅度角度，仍然是一个主要问题。其次，从本地模型更新中提取可靠和干净的成分被证明是具有挑战性的。这种困难源于无法访问与整体训练数据分布密切相关的干净验证数据集，如 [38] 中所述。因此，服务器缺乏用于计算干净模型更新的可信来源。同时，服务器面临着区分良性模型更新和恶意模型更新的困境，使得仅基于本地模型更新计算干净成分成为一项复杂的任务。第三，一个关键方面是确定本地模型更新的适当信用分数。在 RoseAgg 中，所有本地模型更新都被保留而未被移除。然而，如果潜在恶意本地模型更新的累积信用分数很大，则存在损害全局模型完整性的风险。

为了应对上述挑战，RoseAgg 引入了一种多方面的策略。首先，它结合了一种自适应的部分聚合策略，该策略采用归一化、自适应聚类和部分聚合来约束中毒模型更新中观察到的共同特征，例如放大的幅度或相似的定向贡献。此外，RoseAgg 集成了一种降维方法来识别一个可能的干净成分。该成分作为在服务器上对本地模型更新进行自举信用评分的基础。第三，RoseAgg 根据本地模型更新在可能的干净成分上的投影方差为其分配信用评分，确保良性本地模型更新的累积信用评分超过中毒对应物的信用评分。同时，RoseAgg 通过对根据信用评分加权的所有本地模型更新进行平均来计算全局模型更新。

总之，我们的贡献概述如下。

- 我们引入 RoseAgg，一种在存在大量恶意客户端和攻击密度的环境下对定向串通攻击的鲁棒防御。
- 理论上，RoseAgg 对高级定向模型中毒攻击表现出鲁棒性。
- 在四个基准数据集上的广泛评估展示了 RoseAgg 对七种最先进攻击的有效性。

II. 相关工作

A. 联邦学习

在标准的联邦学习场景中，考虑一个由 n 个客户端组成的集合，记为 $C_1, C_2, \dots, C_n$ ，其中每个客户端

$C_i, i \in [1, 2, \dots, n]$ 拥有一个私有的本地数据集 $\mathcal{D}_i$ 。联邦学习中的基本目标是为这些客户端协同获取一个共享的全局模型，而无需将它们的个人数据集传输到中央服务器

[13], [21], [36]。训练过程在多轮中展开。在每一轮中，服务器首先选择一组客户端，并将当前的全局模型发送给他们。然后，每个选中的客户端 C_i 使用其私有数据 $\mathcal{D}_i$ 本地训练模型，并且只有更新的模型参数被发送到中央服务器。最后，这些更新的参数被聚合以生成一个新的全局模型，该模型随后被分发给客户端以进行下一轮训练。这个过程被迭代重复，直到全局模型达到令人满意的性能或完成预定义的轮数。

B. 针对性模型中毒攻击

近期研究 [2], [3], [31], [35], [39] 集中于调查联邦学习中的针对性模型中毒攻击。针对性模型中毒攻击旨在操纵全局训练模型，迫使其对包含预定义攻击者选择模式的任何输入数据预测特定的目标标签 τ （即触发器）。此外，当全局模型遇到未受影响的输入数据时，仍能保持正常行为。形式上，针对性模型中毒攻击可以被视为一个优化问题，如下所示：

$$\begin{aligned} \mathbf{w}^* = \arg \min_{\mathbf{w}} \sum_{i \in \Psi} & \left(\ell \left[f(\mathbf{w}; \mathbf{X}_{cln}^i), \mathbf{Y} \right] \right. \\ & \left. + \ell \left[f(\mathbf{w}; \mathbf{X}_{poi}^i), \mathbf{T} \right] \right) \end{aligned} \quad (1)$$

其中 Ψ 表示被对手 $\mathcal{A}$ 操纵的受影响客户端集合，表示预测标签与真实标签之间的交叉熵损失， $\mathbf{X}_{cln}^i = \{x_1^i, x_2^i, \dots, x_{n_i}^i\}$ 和 $\mathbf{X}_{poi}^i = \{x_1^i, x_2^i, \dots, x_{m_i}^i\}$ 分别代表受影响客户端 i 的干净数据和精心设计的恶意数据， $\mathbf{Y} = \{y_1^i, y_2^i, \dots, y_{n_i}^i\}$ 和 $\mathbf{T} = \{\tau_1^i, \tau_2^i, \dots, \tau_{m_i}^i\}$ 代表干净标签和目标标签。干净标签和攻击者期望的标签。此外，需要注意的是 $\mathbf{X}_{cln}^i \cap \mathbf{X}_{poi}^i = \emptyset$ ，表示恶意数据和干净数据不重叠。针对性模型中毒攻击旨在为给定的恶意数据点 x_j^i 分配最高的目标标签 τ_j 概率，同时同时遵守给定数据点 x_j 的真实标签 y_j 。此外，对手 $\mathcal{A}$ 可以为每个受影响的客户端设计不同的恶意数据，从而进行更隐蔽的攻击。

模型替换攻击（MR）[2] 涉及攻击者 $\mathcal{A}$ 将触发器嵌入到受感染的客户端的本地数据中，并为他们分配攻击者期望的标签。随后， $\mathcal{A}$ 控制受感染客户端的训练过程，生成中毒的本地模型更新。最后， $\mathcal{A}$ 将中毒的本地模型更新以较大的倍数放大，然后提交给服务器。

分布式后门攻击（DBA）[35] 将全局触发模式分解为不同的本地模式，并将它们嵌入到不同受感染方的训练集中。

DBA exhibits greater persistence and stealthiness in federated learning compared to MR.

The label flipping attack (FLIP) [3] aims to cause targeted misclassification of auxiliary data by the global model learned at the server. For instance, in the case of the Fashion-MNIST dataset, the attack aims to misclassify an example originally labeled as class '7' (sneaker) into class '5' (sandal).

The edge-case backdoor attack (EDGE) [31] compels the global model to misclassify seemingly straightforward inputs that are improbable to be part of the training or test data, residing on the tail of the input distribution.

The neurotoxin attack (NEUR) [39] projects the adversarial model updates onto the subspace unused by benign users, thereby enhancing the attack's persistence even after the attacker stops uploading poisoned updates.

The stealthy and colluded attack (CerP) [19] is designed to manipulate FL systems by carefully tuning the collusion between malicious participants. The objective of CerP is to minimize the trigger-induced bias of a poisoned local model compared to a poison-free one. CerP achieves this by jointly adjusting the backdoor trigger and controlling the changes in the poisoned model on each malicious participant. The attack strategy is crafted to be effective against a wide range of state-of-the-art defense mechanisms in FL.

C. Defenses Against Targeted Model Poisoning Attacks

Several recent studies [6], [11], [22], [23] have investigated how to mitigate targeted model poisoning attacks in federated learning.

FoolsGold [11] distinguishes between benign and malicious clients by assessing pairwise similarity among model updates, subsequently down-weighting abnormal updates on the FL server. However, FoolsGold faces challenges in detecting poisoned model updates in the context of “DBA” and multi-objective targeted model poisoning attacks. These attacks generate poisoned model updates without common characteristics, posing a challenge to FoolsGold's ability to distinguish between poisoned and benign updates. Consequently, FoolsGold may exhibit limitations and failures in such scenarios. Moreover, In scenarios where the local data distributions of specific clients exhibit similarity, FoolsGold may erroneously discard the local model updates from these clients due to their similar directional contributions, thereby compromising the overall model performance.

RLR [23] conjectures that the aggregated updates from adversarial and honest agents are likely to differ in the directions, at least for some dimensions. Consequently, they dynamically adjust the learning rate of the aggregation server on a per-dimension and per-round basis, guided the sign information of agents' updates. However, RLR can effectively enhance the robustness of FL systems only if the number of compromised clients by the adversary $\mathcal{A}$ must be significantly lower than a specific threshold value θ .

FLTrust [6] presupposes that the server possesses an additional root dataset for computing a benign model update. Consequently, a local model update receives a lower trust score if its update direction significantly deviates from that of the server model update. This strategy mitigates the impact of

poisoned local updates, effectively thwarting targeted model poisoning attacks. However, the collection of a clean validation dataset poses challenges. FLTrust may falter when there is a significant divergence between the distribution of the validation dataset and the overall training dataset.

FLAME [22] employs model clustering and weight clipping to estimate the requisite level of noise injection, ensuring the eradication of backdoors. It can maintain the benign performance of the aggregated model while effectively eliminating adversarial backdoors. However, when encountering inflation attacks [11] where the proportion of malicious clients is high, FLAME will not provide an effective defense.

Therefore, an additional defense approach is required to enhance the robustness of FL systems.

III. PRELIMINARIES

A. (Density-Based Spatial Clustering of Applications With Noise

DBSCAN (Density-Based Spatial Clustering of Applications with Noise) is a clustering algorithm designed to discover clusters of arbitrary shapes in a dataset. Unlike traditional clustering algorithms that require specifying the number of clusters in advance, DBSCAN identifies clusters based on the density of data points. The algorithm works by selecting a starting data point and expanding the cluster by adding nearby points that have a minimum number of neighbors within a specified radius. Data points that do not meet the density criteria are treated as noise. DBSCAN is particularly effective in handling clusters with irregular shapes and is less sensitive to the order of input data points, making it a valuable tool in various applications, including spatial data analysis and anomaly detection.

B. Principal Component Analysis

The PCA algorithm stands as a foundational technique in the realm of dimensionality reduction. It falls within the domain of linear, unsupervised, and global methods for reducing dimensions in data. Its core principle involves linear mapping, succinctly characterized as the projection of high-dimensional data onto a reduced-dimensional space. In this process, PCA retains crucial principal components, encapsulating substantial data information, while discarding ancillary details of lesser significance for data description. In the context of m data points residing within an n -dimensional space, the following procedure outlines the key steps involved in employing Principal Component Analysis (PCA) for data preprocessing and dimensionality reduction: 1) Data organization: organize the data into an $n \times m$ matrix denoted as X . 2) Zero-centering: perform zero-centering on each row of matrix X by subtracting the mean of each row from all the elements within that row. 3) Covariance matrix computation: calculate the covariance matrix V_m , defined as $V_m = \frac{1}{m}XX^T$. 4) Eigenvalue-eigenvector pair extraction: calculate the eigenvalues and corresponding eigenvectors of the matrix V_m . 5) Eigenvector sorting: arrange the eigenvectors in decreasing order of their corresponding eigenvalues, forming a matrix where each column corresponds to an eigenvector. This matrix

DBA 在联邦学习中表现出比 MR 更强的持久性和隐蔽性

标签翻转攻击 (FLIP) [3] 旨在通过服务器学习的全局模型对辅助数据进行针对性错误分类。例如, 在 Fashion-MNIST 数据集的情况下, 攻击旨在将原本标记为类别 '7' (运动鞋) 的样本错误分类为类别 '5' (凉鞋)。边缘案例后门攻击 (EDGE) [31] 强制全局模型对看似简单的输入进行错误分类, 而这些输入不太可能成为训练或测试数据的一部分, 位于输入分布的尾部。

神经毒素攻击 (NEUR) [39] 将对抗模型更新投影到良性用户未使用的子空间上, 从而在攻击者停止上传中毒更新后增强攻击的持久性。

隐蔽的合谋攻击 (CerP) [19] 旨在通过仔细调整恶意参与者之间的合谋来操纵联邦学习系统。CerP 的目标是使中毒本地模型与无毒模型相比, 在触发诱导的偏差最小化。CerP 通过联合调整后门触发器并控制每个恶意参与者中毒模型的变化来实现这一点。该攻击策略设计为对联邦学习中的多种最先进的防御机制有效。

C. 针对性模型中毒攻击的防御

一些最近的研究 [6], [11], [22], [23] 调查了如何在联邦学习中缓解针对性模型中毒攻击。

FoolsGold [11] 通过评估模型更新之间的成对相似性来区分良性客户端和恶意客户端, 随后在联邦学习服务器上降低异常更新的权重。然而, FoolsGold 在 “DBA” 和多目标针对性模型中毒攻击的背景下检测中毒模型更新时面临挑战。这些攻击生成没有共同特征的中毒模型更新, 这对 FoolsGold 区分中毒和无毒更新的能力构成了挑战。因此, FoolsGold 在这种情况下可能会表现出局限性和失败。此外, 在特定客户端的本地数据分布相似的情况下, FoolsGold 可能由于它们相似的定向贡献而错误地丢弃这些客户端的本地模型更新, 从而影响整体模型性能。

RLR [23] 猜想, 来自敌对和诚实代理的聚合更新在方向上很可能会不同, 至少在某些维度上。因此, 它们根据代理更新符号信息, 动态地按维度和按轮次调整聚合服务器的学习率。然而, 只有当敌对行为者 $\mathcal{A}$ 通过的数量显著低于特定阈值 θ 时, RLR 才能有效地增强 FL 系统的鲁棒性。

FLTrust [6] 假设服务器拥有一个额外的根数据集来计算良性模型更新。因此, 如果本地模型更新的方向与服务器模型更新的方向显著偏差, 则其接收到的信任分数较低。这种策略减轻了

中毒本地更新的影响, 有效地挫败了定向模型中毒攻击。然而, 收集干净的有效数据集提出了挑战。当验证数据集的分布与整体训练数据集的分布之间存在显著差异时, FLTrust 可能会失效。

FLAME [22] 采用模型聚类 and 权重剪裁来估计注入噪声的必要水平, 确保消除后门。它可以保持聚合模型的良性性能, 同时有效消除对抗性后门。然而, 当遇到恶意客户端比例较高的通货膨胀攻击 [11] 时, FLAME 将无法提供有效的防御。

因此, 需要一种额外的防御方法来增强 FL 系统的鲁棒性。

III. PRELIMINARIES

A. (基于密度的空间聚类应用与噪声

DBSCAN (基于密度的空间聚类应用与噪声) 是一种聚类算法, 旨在发现数据集中的任意形状的簇。与需要在事先指定簇数量的传统聚类算法不同, DBSCAN 根据数据点的密度来识别簇。该算法通过选择一个起始数据点, 并通过添加在指定半径内有最小数量邻居的附近点来扩展簇。不满足密度标准的点被视为噪声。DBSCAN 特别适用于处理形状不规则的簇, 并且对输入数据点的顺序不太敏感, 使其成为空间数据分析、异常检测等应用中的有力工具。

B. 主成分分析

PCA 算法是降维领域中的基础技术。它属于线性、无监督和全局的降维方法。其核心原理涉及线性映射, 简明地描述为将高维数据投影到降维空间。在这个过程中, PCA 保留关键的主成分, 封装了大量的数据信息, 同时丢弃了对数据描述意义较小的辅助细节。在 m 个数据点位于 n -维空间的情况下, 以下是采用主成分分析 (PCA) 进行数据预处理和降维的关键步骤: 1) 数据组织: 将数据组织成 $n \times m$ 矩阵, 记为 X 。2) 零中心化: 对矩阵 X 的每一行进行零中心化, 通过从该行的所有元素中减去该行的均值。3) 协方差矩阵计算: 计算协方差矩阵 V_m , 定义为 $V_m = \frac{1}{m}XX^T$ 。4) 特征值 - 特征向量对提取: 计算矩阵 V_m 的特征值和对应的特征向量。5) 特征向量排序: 将特征向量按其对应的特征值降序排列, 形成一个矩阵, 其中每一列对应一个特征向量。这个矩阵

is denoted as P . 6) Dimensionality reduction: the reduced-dimension representation Y is obtained as $Y = P^T X$.

IV. PROBLEM SETUP AND OBJECTIVES

A. Threat Model

This work focuses on defending against targeted model poisoning attacks in federated learning. Without loss of generality, RoseAgg assumes that the data distributed across each client is concealed. We mainly follow the targeted model poisoning attacks in [2], [3], [31], [35], and [39] and establish certain assumptions as outlined below:

- The adversary $\mathcal{A}$ has the same background knowledge of the FL protocol as the benign clients and can observe changes in the global model.
- $\mathcal{A}$ is unable to compromise the server, and it is explicitly forbidden from accessing the training data or local model updates of other honest clients.
- In the scenario where $\mathcal{A}$ only pursues a singular malicious objective, $\mathcal{A}$ can compromise a portion of genuine clients, without imposing any restrictions on the number of compromised clients. Conversely, if $\mathcal{A}$ harbors diverse malicious objectives, it is assumed to have control over no more than half of the clients.

B. Defense Goals

In this work, we aim to develop a robust aggregation rule on the server side, named RoseAgg, to counter targeted collusion attacks. Furthermore, RoseAgg should not sacrifice the utility and efficiency of the system. Concretely, RoseAgg must meet the following objectives:

- *Model performance.* RoseAgg should facilitate the global model in achieving high performance, irrespective of the presence of attacks.
- *Robustness.* RoseAgg must thwart the adversary from achieving its malicious objectives. This involves mitigating or eliminating the impact of poisoned model updates, which is crucial for ensuring that the aggregated global model remains free from any attacker-chosen malicious behavior.
- *Efficiency.* RoseAgg should not introduce excessive communication and computation overhead compared.

V. ROSEAGG OVERVIEW AND DESIGN

In this section, we meticulously elucidate the high-level idea of RoseAgg along with its detailed designs. Moreover, for clarity and ease of comprehension, Table I provides a comprehensive list of symbols employed in this work along with their corresponding explanations.

A. High-Level Idea

1) *Motivation:* State-of-the-art work [6] illustrates that despite the effectiveness of recent defenses [4], [8], [12], [37], the adversary can still corrupt the global model by submitting carefully crafted model updates to the server. The fundamental reason behind this is that the server lacks a trusted source and therefore cannot distinguish the poisoned local updates.

TABLE I
NOTATIONS AND CORRESPONDING DEFINITIONS

Notation	Definition
n	The number of the total clients.
$\mathcal{M}$	The number of the participants during each round.
$\mathcal{D}_i$	The private local dataset of client C_i .
$\mathbf{w}_t$	The global model at round t .
$\mathbf{g}_i^t$	The local update of client C_i during the t -th round.
$\bar{\mathbf{g}}_i^t$	The normalized local update of client C_i at round t .
$\mathcal{C}_i^t$	The i -th cluster at round t .
$\bar{\mathbf{g}}_j^{t,i}$	The j -th normalized local update with in i -th cluster during the t -th round.
$\tilde{\mathbf{g}}^{t,i}$	The partial aggregated update of the i -th cluster during the t -th round.
$\tilde{\mathbf{g}}_0^t$	The clean ingredient at round t .
$\mathbf{g}^t$	The global update at round t .
$\bar{\mathbf{g}}^t$	The normalized global update at round t .
$F(\mathbf{w})$	The expected empirical loss function concerning the model $\mathbf{w}$ over $\{\mathcal{D}_i\}_{i=1}^n$.
$f(\mathcal{D}, \mathbf{w})$	The empirical loss function concerning the model $\mathbf{w}$ on $\mathcal{D}$.

To bridge the gap, they propose FLTrust [6], a trust-bootstrap-based FL framework for mitigating byzantine and targeted model poisoning attacks. FLTrust assumes that the server can acquire a clean root dataset for computing a benign model update. The server then assigns a trust score to each participant according to the direction deviation between the local model update and the server model update. However, this proposition encounters challenges in typical FL settings. Firstly, the server might encounter incomplete data collection, such as missing data from a specific class. Secondly, there's a risk of collecting poisoned data during the data collection process. In summary, these potential biases or inaccuracies may harm the robustness and reliability of the algorithm. Therefore, a generic method is required to compute a plausible clean model update solely based on the received local model updates, and to assign reasonable weights to these updates for aggregation.

2) *RoseAgg Overview:* During each communication round between the server and participants, RoseAgg follows the general steps of FL discussed in Section II. In general, the adversary $\mathcal{A}$ has the ability to manipulate the directions and the magnitudes of local model updates on malicious participants, which can cause the global model to skew in the poisoned direction, resulting in targeted model poisoning attacks [22]. Therefore, RoseAgg considers both the directions and the magnitudes of local model updates when designing our defense strategy. Firstly, RoseAgg employs an adaptive partial aggregating process to mitigate the influence of local updates exhibiting scaled-up magnitudes or similar directional contributions. Secondly, RoseAgg calculates and maintains a plausible clean ingredient on the server, which is utilized to bootstrap credit scores for model updates. Thirdly, RoseAgg assigns credit scores to local model updates based on their projection variances on the plausible clean ingredient. Finally, RoseAgg computes a global model update by averaging local model updates weighted by the credit scores, thereby updating the global model.

被表示为 P 。6) 降维：降维表示 Y 是作为 $Y = P^T X$ 获得的。

IV. 问题设置和目标

A. 威胁模型

这项工作专注于防御联邦学习中的定向模型中毒攻击。不失一般性，RoseAgg 假设每个客户端上分布的数据是保密的。我们主要遵循 中的定向模型中毒攻击，并建立以下假设：

- 攻击者 $\mathcal{A}$ 拥有与良性客户端相同的 FL 协议背景知识，并且可以观察到全局模型的变化。
- $\mathcal{A}$ 无法破坏服务器，并且明确禁止访问其他诚实客户端的训练数据或本地模型更新。
- 在 $\mathcal{A}$ 仅追求单一恶意目标的情况下， $\mathcal{A}$ 可以破坏部分真实客户端，而不对被破坏客户端的数量施加任何限制。相反，如果 $\mathcal{A}$ 怀有多个恶意目标，则假设其控制不超过一半的客户端。

B. 防御目标

在这项工作中，我们旨在在服务器端开发一种鲁棒的聚合规则，名为 RoseAgg，以对抗定向共谋攻击。此外，RoseAgg 不应牺牲系统的效用和效率。具体来说，RoseAgg 必须满足以下目标：

- 模型性能。RoseAgg 应促进全局模型在存在攻击的情况下实现高性能。
- 鲁棒性。RoseAgg 必须挫败对手实现其恶意目标。这涉及减轻或消除污染模型更新的影响，这对于确保聚合的全局模型不受任何攻击者选择的恶意行为至关重要。

- 效率。RoseAgg 不应引入过多的通信和计算开销。

V. ROSEAGG 概述和设计

在本节中，我们详细阐述了 RoseAgg 的高层次想法及其详细设计。此外，为了清晰和易于理解，表 I 提供了本工作中使用的符号的综合列表及其相应的解释。

A. 高层次想法

1) 动机：最先进的工作 [6] 说明，尽管最近的防御 [4], [8], [12], [37], 有效，但对手仍然可以通过向服务器提交精心设计的模型更新来破坏全局模型。其根本原因是服务器缺乏可信来源，因此无法区分污染的本地更新。

表 I 符号及其对应定义

Notation	Definition
n	The number of the total clients.
$\mathcal{M}$	The number of the participants during each round.
$\mathcal{D}_i$	The private local dataset of client C_i .
$\mathbf{w}_t$	The global model at round t .
$\mathbf{g}_i^t$	The local update of client C_i during the t -th round.
$\bar{\mathbf{g}}_i^t$	The normalized local update of client C_i at round t .
$\mathcal{C}_i^t$	The i -th cluster at round t .
$\bar{\mathbf{g}}_j^{t,i}$	The j -th normalized local update with in i -th cluster during the t -th round.
$\tilde{\mathbf{g}}^{t,i}$	The partial aggregated update of the i -th cluster during the t -th round.
$\tilde{\mathbf{g}}_0^t$	The clean ingredient at round t .
$\mathbf{g}^t$	The global update at round t .
$\bar{\mathbf{g}}^t$	The normalized global update at round t .
$F(\mathbf{w})$	The expected empirical loss function concerning the model $\mathbf{w}$ over $\{\mathcal{D}_i\}_{i=1}^n$.
$f(\mathcal{D}, \mathbf{w})$	The empirical loss function concerning the model $\mathbf{w}$ on $\mathcal{D}$.

为弥补这一差距，他们提出了基于信任引导的 FL 框架 FLTrust [6], 用于减轻拜占庭攻击和针对性模型中毒攻击。FLTrust 假设服务器可以获得一个干净的根数据集，用于计算良性模型更新。然后，服务器根据本地模型更新与服务器模型更新之间的方向偏差，为每个参与者分配一个信任分数。然而，这一提议在典型的联邦学习设置中遇到了挑战。首先，服务器可能会遇到不完整的数据收集，例如特定类别中的缺失数据。其次，在数据收集过程中存在收集中毒数据的风险。总之，这些潜在的偏差或不准确性可能会损害算法的鲁棒性和可靠性。因此，需要一种通用方法，仅基于收到的本地模型更新来计算一个合理的干净模型更新，并为这些更新分配合理的权重进行聚合。

2) *RoseAgg 概述:* 在服务器和参与者之间的每次通信轮次中，RoseAgg 遵循第 II 节中讨论的联邦学习的一般步骤。通常，攻击者 $\mathcal{A}$ 有能力操纵恶意参与者本地模型更新的方向和幅度，这会导致全局模型在中毒方向上倾斜，从而造成针对性模型中毒攻击 [22]。因此，RoseAgg 在设计我们的防御策略时考虑了本地模型更新的方向和幅度。首先，RoseAgg 采用自适应部分聚合过程，以减轻表现出大幅度或类似方向贡献的本地更新的影响。其次，RoseAgg 计算并维护服务器上的一个合理的干净成分，用于为模型更新引导信用分数。第三，RoseAgg 根据本地模型更新在其对合理干净成分上的投影方差为其分配信用分数。最后，RoseAgg 通过对信用分数加权的本地模型更新进行平均，计算全局模型更新，从而更新全局模型。

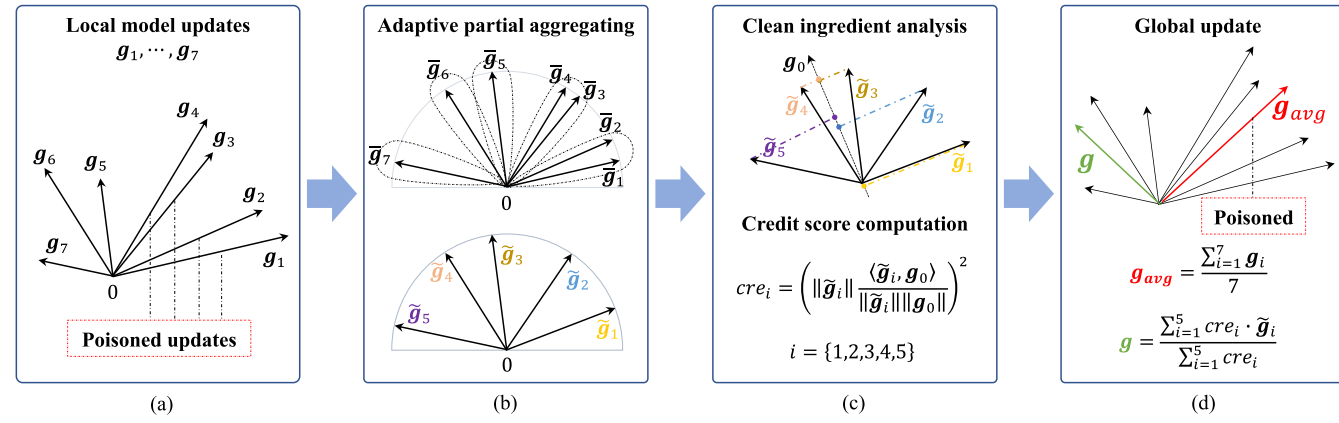


Fig. 2. Illustration of RoseAgg, which is applied during each communication round between the server and participants. Notably, g_{avg} and g represent global updates computed using the methods of simple average and RoseAgg, respectively.

Algorithm 1 LocalUpdate

Input: $w, \mathcal{D}, b, \eta, E_l$ $\triangleright E_l$ is the local epochs, η is the local learning rate, and b is the batch size

Output: Local model update.

- 1: $w^0 \leftarrow w$
- 2: **for** e in $[1, E_l]$ **do**
- 3: Randomly sample a batch $\mathcal{D}_b$ with the size of b from $\mathcal{D}$.
- 4: $w^e \leftarrow w^{e-1} - \eta \nabla f(\mathcal{D}_b, w^{e-1})$.
- 5: **end for**
- 6: **return** $w^{E_l} - w^0$

B. RoseAgg Design

As described in Section V-A, RoseAgg consists of three main components: adaptive partial aggregating, clean ingredient extraction, and credit score computation. Figure 2 illustrates the corresponding components and the workflow of RoseAgg. Algorithm 2 outlines the procedure of RoseAgg.

1) *Adaptive Partial Aggregating*: The adaptive partial aggregating component consists of three parts: normalizing, adaptive clustering, and partial aggregating. The adaptive partial aggregating step is shown in lines 7-11 of Algorithm 2.

a) *Normalizing*: The adversary can scale up the magnitude of poisoned local model updates [2], [35], which amplifies their impact on the global model update and enables successfully targeted model poisoning. To counter this, RoseAgg uses normalization to enable all local model updates to lie in the same hyper-sphere. Namely, RoseAgg only considers the direction of each local update in this stage. Formally:

$$\bar{g}_i^t = \frac{g_i^t}{\|g_i^t\|} \quad (2)$$

The normalization prevents any single local update from excessively impacting the global model update. Moreover, it can also eliminate bias stemming from variations in the magnitude of local updates, which may arise when employing a dimension-reduction method to identify a plausible clean ingredient.

b) *Dynamic clustering*: The technique of Dynamic clustering, based on DBSCAN [10], identifies local model updates

Algorithm 2 RoseAgg

Input: n, w_0, T $\triangleright w_0$ is the initial global model, T is the total communication rounds.

Output: w_T $\triangleright w_T$ is the global model after T rounds.

- 1: **for** each round t in $[1, T]$ **do**
- 2: The server randomly samples $\mathcal{M}$ participants.
- 3: **for** each client C_i in $[C_1, C_2, \dots, C_{\mathcal{M}}]$ **parallel do**
- 4: $g_i^t \leftarrow \text{LocalUpdate}(w_{t-1}, \mathcal{D}_i, b, \eta, E_l)$.
- 5: Submit g_i^t to the server.
- 6: **end for**
- 7: $\bar{g}_i^t \leftarrow \frac{g_i^t}{\|g_i^t\|}, \forall i \in [1, \mathcal{M}]$ $\triangleright$ Normalizing
- 8: $(C_1^t, \dots, C_{K_t}^t) \leftarrow \text{Clustering}(g_1^t, g_2^t, \dots, g_{\mathcal{M}}^t)$. $\triangleright$ Dynamically clustering local updates to K_t clusters.
- 9: **for** i in $[1, K_t]$ **do**
- 10: Calculate $\tilde{g}^{t,i}$. $\triangleright$ Partial aggregating.
- 11: **end for**
- 12: Stacking $\{\tilde{g}^{t,i}\}_{i=1}^{K_t}$ to form a matrix $\mathcal{H}^t$ $\triangleright \mathcal{H}^t \in \mathbb{R}^{d \times K_t}$, d is the dimension of each model update.
- 13: Renormalizing each column of $\mathcal{H}^t$.
- 14: $\hat{\mathcal{H}}^t \leftarrow \mathcal{H}^{tT} \mathcal{H}^t$
- 15: $\bar{g}_0^t \leftarrow \mathcal{H}^t \xi_{max} / \sqrt{\lambda_{max}}$, λ_{max} and ξ_{max} are the maximum eigenvalue and corresponding eigenvector of $\hat{\mathcal{H}}^t$.
- 16: $cre_i^t \leftarrow \left(\|\tilde{g}^{t,i}\| \frac{\langle \tilde{g}^{t,i}, \bar{g}_0^t \rangle}{\|\tilde{g}^{t,i}\| \|\bar{g}_0^t\|} \right)^2$, $g^t \leftarrow \frac{\sum_i cre_i^t \cdot \tilde{g}^{t,i}}{\sum_i cre_i^t}, \forall i \in [1, K_t]$, $g^t \leftarrow \gamma_t \frac{g^t}{\|g^t\|}$ $\triangleright \gamma_t$ relies on the magnitudes of local updates at round t .
- 17: $w_t \leftarrow w_{t-1} - g^t$.
- 18: **end for**

that have similar direction contributions. The state-of-the-art defense [22] clusters local models based on pairwise cosine distances. They set the minimum cluster size to be $\lfloor \frac{n}{2} \rfloor + 1$ so that the resulting cluster contains the majority of the local models. The remaining local models are considered potentially malicious and are therefore removed. In addition, the defenses [26], [38] use K-means to cluster all local model updates into two groups, retaining only the larger group for aggregation. However, when the proportion of malicious

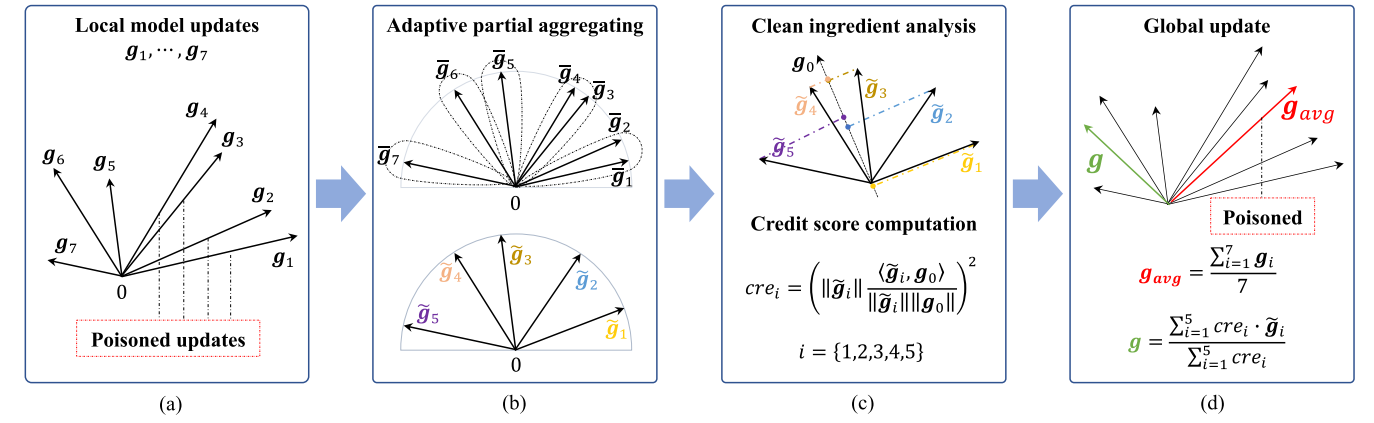


Fig. 2. RoseAgg 的示意图，该图应用于服务器与参与者之间每次通信轮次。值得注意的是，使用简单平均方法和 RoseAgg 方法分别计算的 g_{avg} 和 g 代表全局更新。

算法 1 LocalUpdate

输入: $w, \mathcal{D}, b, \eta, E_l$ $\triangleright E_l$ 是本地轮数, η 是本地学习率, 以及 b 是批量大小

输出: 本地模型更新。

- 1: $w^0 \leftarrow w$
- 2: **for** e in $[1, E_l]$ **do**
- 3: 随机采样一个大小为 $\mathcal{D}_b$ 的批次 b 从 $\mathcal{D}$.
- 4: $w^e \leftarrow w^{e-1} - \eta \nabla f(\mathcal{D}_b, w^{e-1})$.
- 5: **endfor**
- 6: **return** $w^{E_l} - w^0$

B. RoseAgg 设计

如第 V-A 节所述, RoseAgg 由三个主要组件组成: 自适应部分聚合、清洁成分提取和信用评分计算。图 2 展示了相应的组件和 RoseAgg 的工作流程。算法 2 概述了 RoseAgg 的步骤。

1) 自适应部分聚合: 自适应部分聚合组件由三个部分组成: 归一化、自适应聚类和部分聚合。自适应部分聚合步骤在算法 2 的行 7-11 中显示。

a) 归一化: 攻击者可以放大中毒本地模型更新的幅度 [2], [35], 从而增强其对全局模型更新的影响并成功进行模型中毒。为应对此问题, RoseAgg 使用归一化方法, 使所有本地模型更新都位于同一超球体内。具体来说, RoseAgg 在此阶段仅考虑每个本地更新的方向。形式上:

$$\bar{g}_i^t = \frac{g_i^t}{\|g_i^t\|} \quad (2)$$

归一化可以防止任何单个本地更新过度影响全局模型更新。此外, 它还可以消除本地更新幅度变化引起的偏差, 这种变化可能在使用降维方法识别可能的干净成分时出现。

b) 动态聚类: 动态聚类技术, 基于 DBSCAN [10], 识别具有相似方向贡献的本地模型更新

算法 2 RoseAgg

输入: n, w_0, T $\triangleright w_0$ 是初始全局模型, T 是总通信轮数。

输出: w_T $\triangleright w_T$ 是全局模型 after T rounds.

- 1: **for** each round t in $[1, T]$ **do**
- 2: The server randomly samples $\mathcal{M}$ participants.
- 3: **for** each client C_i in $[C_1, C_2, \dots, C_{\mathcal{M}}]$ **parallel do**
- 4: $g_i^t \leftarrow \text{LocalUpdate}(w_{t-1}, \mathcal{D}_i, b, \eta, E_l)$.
- 5: Submit g_i^t to the server.
- 6: **end for**
- 7: $\bar{g}_i^t \leftarrow \frac{g_i^t}{\|g_i^t\|}, \forall i \in [1, \mathcal{M}]$ $\triangleright$ Normalizing
- 8: $(C_1^t, \dots, C_{K_t}^t) \leftarrow \text{Clustering}(g_1^t, g_2^t, \dots, g_{\mathcal{M}}^t)$. $\triangleright$ Dynamically clustering local updates to K_t clusters.
- 9: **for** i in $[1, K_t]$ **do**
- 10: Calculate $\tilde{g}^{t,i}$. $\triangleright$ Partial aggregating.
- 11: **end for**
- 12: Stacking $\{\tilde{g}^{t,i}\}_{i=1}^{K_t}$ to form a matrix $\mathcal{H}^t$ $\triangleright \mathcal{H}^t \in \mathbb{R}^{d \times K_t}$, d is the dimension of each model update.
- 13: Renormalizing each column of $\mathcal{H}^t$.
- 14: $\hat{\mathcal{H}}^t \leftarrow \mathcal{H}^{tT} \mathcal{H}^t$
- 15: $\bar{g}_0^t \leftarrow \mathcal{H}^t \xi_{max} / \sqrt{\lambda_{max}}$, λ_{max} and ξ_{max} are the maximum eigenvalue and corresponding eigenvector of $\hat{\mathcal{H}}^t$.
- 16: $cre_i^t \leftarrow \left(\|\tilde{g}^{t,i}\| \frac{\langle \tilde{g}^{t,i}, \bar{g}_0^t \rangle}{\|\tilde{g}^{t,i}\| \|\bar{g}_0^t\|} \right)^2$, $g^t \leftarrow \frac{\sum_i cre_i^t \cdot \tilde{g}^{t,i}}{\sum_i cre_i^t}, \forall i \in [1, K_t]$, $g^t \leftarrow \gamma_t \frac{g^t}{\|g^t\|}$ $\triangleright \gamma_t$ relies on the magnitudes of local updates at round t .
- 17: $w_t \leftarrow w_{t-1} - g^t$.
- 18: **end for**

这些本地模型更新具有相似的方向贡献。最先进的防御 [22] 基于成对余弦距离对本地模型进行聚类。他们将最小聚类大小设置为 $\lfloor \frac{n}{2} \rfloor + 1$, 以便生成的聚类包含大多数本地模型。其余的本地模型被认为是潜在的恶意模型, 因此被移除。此外, 防御 [26], [38] 使用 K-means 将所有本地模型更新聚类为两个组, 仅保留较大的组进行聚合。然而, 当恶意模型的

clients is high, these approaches may retain the poisoned models after clustering, increasing the impact of poisoned local model updates.

FL operates on a decentralized model, with local models trained on data distributed across multiple clients or nodes. These clients can have variations in data distribution, data quality, or even the nature of the data itself. This inherent data heterogeneity can lead to diverse local models. Therefore, even if some local models appear to be outliers or noisy (meaning they might not perform as well as others or seem different from the majority), they should not be discarded or removed during the process of aggregating these models to form a global model. The reason for not removing these diverse local models is that they might still contain valuable information. They could be representative of unique but relevant data patterns that are specific to that particular client. By including these diverse models in the aggregation process, the global model can potentially learn from this variety and become more robust and adaptable. Consequently, when applying the DBSCAN algorithm for clustering, the parameter for the minimum number of samples required to form a cluster is set to **1**. This means that even a single model can be considered a separate cluster in this context. In simple terms, when RoseAgg uses the DBSCAN algorithm, it doesn't identify any of the local models as noise. All models are treated as potentially meaningful and are assigned to clusters.

c) *Partial aggregating*: As the local model updates within each cluster have similar direction contributions, RoseAgg computes a model update for each cluster to represent its directional contribution. Such a way can eliminate the redundancy of similar directional contributions, therefore limiting the impact of potentially malicious model updates from the directional perspective. Specifically, a partial aggregating module is employed to achieve this. Assuming the i -th cluster contains κ local updates during the round t , forming a distance square matrix $S^{t,i}$ of size $\kappa \times \kappa$, i.e., $S^{t,i} \in \mathbb{R}^{\kappa \times \kappa}$. The partially aggregated update $\tilde{\mathbf{g}}^{t,i}$ is calculated as follows:

$$\begin{cases} \tilde{\mathbf{g}}^{t,i} = \bar{\mathbf{g}}_{\kappa}^{t,i}, & \kappa = 1 \\ \tilde{\mathbf{g}}^{t,i} = \frac{\sum_{j=1}^{\kappa} \alpha_j^{t,i} \cdot \bar{\mathbf{g}}_j^{t,i}}{\sum_{j=1}^{\kappa} \alpha_j^{t,i}}. & \kappa > 1 \end{cases} \quad (3)$$

where $\alpha_j^{t,i}$ represents the summation of the j -th row from the distance matrix $S^{t,i}$.

Before extracting the clean ingredient of the local model updates, $\tilde{\mathbf{g}}^{t,i}$ need to be re-normalized to avoid the influence of the magnitude of the model updates on the extraction result. The reason for re-normalization is that the process of partial aggregating involves a weighted sum of vectors such that the resulting $\tilde{\mathbf{g}}_i^t$ has a magnitude less than 1.

2) *Clean Ingredient Extraction*: The clean ingredient denotes a reliable ingredient extracted from local model updates, satisfying the condition where the sum of projection variances of benign updates onto it surpasses the sum of projection variances of poisoned updates onto the same ingredient. Suppose that all local updates are organized into K_t clusters during the adaptive partial aggregating phase at round t . Formally, $\tilde{\mathbf{g}}_0^t$ is considered as the clean ingredient

if the following inequality holds:

$$\sum_{i=1}^m \left(\|\tilde{\mathbf{g}}^{t,i}\| \frac{\langle \tilde{\mathbf{g}}^{t,i}, \tilde{\mathbf{g}}_0^t \rangle}{\|\tilde{\mathbf{g}}^{t,i}\| \|\tilde{\mathbf{g}}_0^t\|} \right)^2 < \sum_{i=m+1}^{K_t} \left(\|\tilde{\mathbf{g}}^{t,i}\| \frac{\langle \tilde{\mathbf{g}}^{t,i}, \tilde{\mathbf{g}}_0^t \rangle}{\|\tilde{\mathbf{g}}^{t,i}\| \|\tilde{\mathbf{g}}_0^t\|} \right)^2 \quad (4)$$

where $\langle \cdot \rangle$ represents the dot product of two vectors, clusters $1, \dots, m$ encompass poisoned updates, referred to “poisoned clusters”, while clusters $m+1, \dots, K_t$ do not include poisoned updates, referred to “benign clusters”.

Figure 3 displays the pairwise cosine distance among local model updates. It is noticeable that during targeted model poisoning attacks, if attackers have a shared malicious goal, their model updates exhibit more pronounced similarities than those among most benign clients. Our hypothesis is that, since attackers have a shared malicious goal, the feature spaces relevant to the malicious goal would be very similar for each attacker. The similarity of the feature spaces can be measured using the pairwise cosine distance of the local model updates. Moreover, clients who have similar local data distribution would also exhibit some similarities. Therefore, it is essential to retain all clusters as the clustering process cannot differentiate which cluster contains attackers.

Based on specific attack assumptions illustrated in Section IV-A, the adaptive partial aggregating module enables poisoned updates to be organized into relatively few clusters. Concretely, the adaptive partial aggregating module in RoseAgg groups local updates into clusters, ensuring the number of potential “benign clusters” surpasses that of potential “poisoned clusters”. Consequently, when represented model updates are extracted from these clusters, the number of potential benign updates surpasses that of potential poisoned updates. Furthermore, RoseAgg utilizes the concept of maximizing variance via Principal Component Analysis (PCA) to isolate a principal component. Given that all updates are standardized and benign updates outnumber poisoned updates, it is self-evident that the sum of projected variances of benign updates on this principal component is greater than the sum of projected variances of poisoned updates on the same component. As a result, this principal component is considered benign.

Concretely, each partially aggregated model update, denoted as $\tilde{\mathbf{g}}^{t,i}$, is viewed as a high-dimensional data point. Then the collection $(\tilde{\mathbf{g}}^{t,1}, \tilde{\mathbf{g}}^{t,2}, \dots, \tilde{\mathbf{g}}^{t,K_t})$ can be regard as a compact dataset, comprising K_t data points. Each of these data points is d -dimensional, where d is the dimension of each local model update. To formalize this dataset, RoseAgg employs a matrix $\mathcal{H}^t \in \mathbb{R}^{d \times K_t}$. The goal is to extract the first principal component of the matrix $\mathcal{H}^t$ using PCA [32]. This principal component serves as the desired ingredient.

The matrix $\mathcal{H}^t$ has column vectors with very large dimensions. Using PCA directly would demand massive computation and memory resources. Thus, the desired ingredient is extracted using the following approach:

$$\begin{aligned} \hat{\mathcal{H}}^t &\leftarrow \mathcal{H}^{tT} \mathcal{H}^t \\ \tilde{\mathbf{g}}_0^t &\leftarrow \mathcal{H}^t \xi_{\max} / \sqrt{\lambda_{\max}} \end{aligned} \quad (5)$$

客户端数量较多, 这些方法在聚类后可能仍然保留中毒模型, 增加中毒本地模型更新的影响。

联邦学习 (FL) 在去中心化模型上运行, 本地模型在分布在多个客户端或节点上的数据上训练。这些客户端在数据分布、数据质量或甚至数据本身的性质上可能存在差异。这种固有的数据异构性可能导致不同的本地模型。因此, 即使某些本地模型看起来是异常值或噪声 (这意味着它们可能不如其他模型表现好或与大多数模型不同), 在聚合这些模型以形成全局模型的过程中也不应丢弃或删除。不移除这些多样化的本地模型的原因是它们可能仍然包含有价值的信息。它们可能代表了特定客户端特有的但相关的数据模式。通过将这些多样化的模型包含在聚合过程中, 全局模型可以潜在地从这种多样性中学习, 并变得更加鲁棒和适应性强。因此, 在应用 DBSCAN 算法进行聚类时, 形成集群所需的最小样本数参数设置为 **1**。这意味着在这种情况下, 即使单个模型也可以被视为一个单独的集群。简单来说, 当 RoseAgg 使用 DBSCAN 算法时, 它不会将任何本地模型识别为噪声。所有模型都被视为潜在的、有意义的, 并被分配到集群中。

c) 部分聚合: 由于每个集群内的本地模型更新具有相似的贡献方向, RoseAgg 为每个集群计算一个模型更新来表示其方向贡献。这种方法可以消除相似方向贡献的冗余, 因此从方向角度限制潜在恶意模型更新的影响。具体来说, 采用一个部分聚合模块来实现这一点。假设第 i -个集群在轮次 t 中包含 κ 个本地更新, 形成一个大小为 $\kappa \times \kappa$ 的距离平方矩阵 $S^{t,i}$, 即 $i, S^{t,i} \in \mathbb{R}^{\kappa \times \kappa}$ 。部分聚合更新 $\tilde{\mathbf{g}}^{t,i}$ 的计算方法如下:

$$\begin{cases} \tilde{\mathbf{g}}^{t,i} = \bar{\mathbf{g}}_{\kappa}^{t,i}, & \kappa = 1 \\ \tilde{\mathbf{g}}^{t,i} = \frac{\sum_{j=1}^{\kappa} \alpha_j^{t,i} \cdot \bar{\mathbf{g}}_j^{t,i}}{\sum_{j=1}^{\kappa} \alpha_j^{t,i}}. & \kappa > 1 \end{cases} \quad (3)$$

其中 $\alpha_{j,i}^t$ 表示距离矩阵 $S^{t,i}$ 的第 j 行的总和。

在提取本地模型更新的干净成分之前, $\tilde{\mathbf{g}}^{t,i}$ 需要重新归一化, 以避免模型更新幅度对提取结果的影响。重新归一化的原因是部分聚合过程涉及向量的加权求和, 因此得到的 $\tilde{\mathbf{g}}_i^t$ 的幅度小于 1。

2) 干净成分提取: 干净成分指的是从本地模型更新中提取的可靠成分, 满足良性更新在该成分上的投影方差之和超过中毒更新在该成分上的投影方差之和的条件。假设在轮次 t 的自适应部分聚合阶段, 所有本地更新被组织成 K_t 个簇。形式上, $\tilde{\mathbf{g}}_0^t$ 被视为干净成分

如果满足以下不等式:

$$\sum_{i=1}^m \left(\|\tilde{\mathbf{g}}^{t,i}\| \frac{\langle \tilde{\mathbf{g}}^{t,i}, \tilde{\mathbf{g}}_0^t \rangle}{\|\tilde{\mathbf{g}}^{t,i}\| \|\tilde{\mathbf{g}}_0^t\|} \right)^2 < \sum_{i=m+1}^{K_t} \left(\|\tilde{\mathbf{g}}^{t,i}\| \frac{\langle \tilde{\mathbf{g}}^{t,i}, \tilde{\mathbf{g}}_0^t \rangle}{\|\tilde{\mathbf{g}}^{t,i}\| \|\tilde{\mathbf{g}}_0^t\|} \right)^2 \quad (4)$$

其中 $\langle \cdot \rangle$ 表示两个向量的点积, 簇 $1, \dots, m$ 包含中毒更新, 称为 “中毒簇”, 而簇 $m+1, \dots, K_t$ 不包含中毒更新, 称为 “良性簇”。

图 3 显示了本地模型更新之间的成对余弦距离。值得注意的是, 在定向模型中毒攻击中, 如果攻击者有共同的恶意目标, 他们的模型更新比大多数良性客户端之间的更新表现出更明显的相似性。我们的假设是, 由于攻击者有共同的恶意目标, 每个攻击者相关的恶意目标特征空间会非常相似。特征空间的相似性可以使用本地模型更新的成对余弦距离来衡量。此外, 具有相似本地数据分布的客户端也会表现出一些相似性。因此, 必须保留所有簇, 因为聚类过程无法区分哪个簇包含攻击者。

基于第四节 A 中阐述的特定攻击假设, 自适应部分聚合模块能够将污染更新组织成相对较少的簇。具体而言, RoseAgg 中的自适应部分聚合模块将本地更新分组为簇, 确保潜在 “良性簇” 的数量超过潜在 “污染簇” 的数量。因此, 当从这些簇中提取表示模型更新时, 潜在良性更新的数量将超过潜在污染更新的数量。此外, RoseAgg 利用主成分分析 (PCA) 最大化方差的概念来分离一个主成分。鉴于所有更新都经过标准化且良性更新数量超过污染更新数量, 显然良性更新在这个主成分上的投影方差之和将大于污染更新在同一主成分上的投影方差之和。因此, 该主成分被认为是良性的。

具体而言, 每个部分聚合的模型更新, 记为 $\tilde{\mathbf{g}}^{t,i}$, 被视为一个高维数据点。然后集合 $(\tilde{\mathbf{g}}^{t,1}, \tilde{\mathbf{g}}^{t,2}, \dots, \tilde{\mathbf{g}}^{t,K_t})$

可以被视为一个紧凑的数据集, 包含 K_t 个数据点。这些数据点中的每一个是 d - 维的, 其中 d 是每个本地模型更新的维度。为了形式化这个数据集, RoseAgg 采用矩阵 $\mathcal{H}^t \in \mathbb{R}^{d \times K_t}$ 。目标是通过 PCA [32] 提取矩阵 $\mathcal{H}^t$ 的第一主成分。该主成分即为所需的成分。

矩阵 $\mathcal{H}^t$ 具有非常大的列向量维度。直接使用 PCA 会需要大量的计算和内存资源。因此, 使用以下方法提取所需的成分:

$$\begin{aligned} \hat{\mathcal{H}}^t &\leftarrow \mathcal{H}^{tT} \mathcal{H}^t \\ \tilde{\mathbf{g}}_0^t &\leftarrow \mathcal{H}^t \xi_{\max} / \sqrt{\lambda_{\max}} \end{aligned} \quad (5)$$

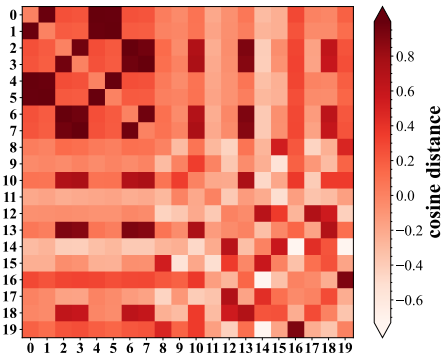


Fig. 3. Cosine distance matrix of local updates. The matrix includes client IDs ranging from 0 to 19. The objectives of Clients 0 and 4 are to misclassify images containing a pixel pattern “F” as airplanes. Clients 1 and 5, on the other hand, intend to misidentify images with a pixel pattern “L” as airplanes. Clients 2 and 6 specifically aim to misclassify images featuring a pixel pattern resembling the letter “A” as birds. Lastly, Clients 3 and 7 focus on misidentifying images of cars that have vertically striped walls in the background as birds.

where $\tilde{\mathbf{g}}_0^t$ is the desired ingredient, $\lambda_{\max}$ and $\xi_{\max}$ are the maximum eigenvalue and corresponding eigenvector of $\hat{\mathbf{H}}^t$. In addition, since $\hat{\mathbf{H}}^t \in \mathbb{R}^{K_t \times K_t}$ is a low-dimensional matrix, the computation overhead for calculating $\lambda_{\max}$ and $\xi_{\max}$ on the server would not be excessive. The clean ingredient extraction step is shown in lines 12-15 of Algorithm 2.

3) *Credit Score Computation*: FLTrust [6] assigns each client a trust score based on the ReLU-clipped cosine distance. However, this could assign the model update of 0 with large angular between the model update maintained by the server. Namely, FLTrust considers such model updates as malicious updates and excludes them. Nevertheless, such angular deviation may be caused by the data heterogeneity [14], [16], [24]. Simply removing the model updates may cause accuracy deterioration of the global model. Therefore, an alternative approach is required.

Specifically, RoseAgg uses the projection variance of each partially aggregated update on the clean ingredient as its credit score. Formally, the credit score is defined as follows:

$$cre_i^t \leftarrow \left(\left\| \tilde{\mathbf{g}}^{t,i} \right\| \frac{\langle \tilde{\mathbf{g}}^{t,i}, \tilde{\mathbf{g}}_0^t \rangle}{\left\| \tilde{\mathbf{g}}^{t,i} \right\| \left\| \tilde{\mathbf{g}}_0^t \right\|} \right)^2, \quad i \in [1, K_t] \quad (6)$$

where cre_i^t is the credit score for the i -th partially aggregated update $\tilde{\mathbf{g}}^{t,i}$ at round t . The rationale for calculating credit scores in this manner is based on the fact that $\tilde{\mathbf{g}}_0^t$ is a plausible clean ingredient. Consequently, the summation of projection variances of benign data points will be greater than that of poisoned data points. As a result, the weight of benign data points will be greater than that of poisoned data points in weighted summation, thereby mitigating model poisoning attacks.

4) *Updating the Global Model*: The global update is calculated by averaging the partially aggregated model updates weighted by the credit scores:

$$\mathbf{g}^t \leftarrow \frac{\sum_{i=1}^{K_t} cre_i^t \tilde{\mathbf{g}}^{t,i}}{\sum_{i=1}^{K_t} cre_i^t}$$

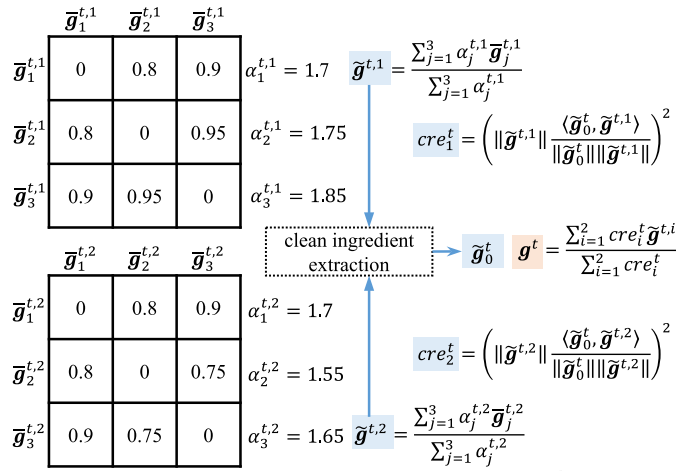


Fig. 4. Toy example of RoseAgg methodology.

$$\tilde{\mathbf{g}}^t \leftarrow \frac{\mathbf{g}^t}{\left\| \mathbf{g}^t \right\|} \quad (7)$$

where $\tilde{\mathbf{g}}^t$ is the normalized global update at round t . In this way, RoseAgg can incorporate the contributions of all participants while giving more weight to benign participants. Subsequently, the server proceeds to update the global model as follows:

$$\mathbf{w}_t \leftarrow \mathbf{w}_{t-1} - \gamma_t \tilde{\mathbf{g}}^t \quad (8)$$

where $\mathbf{w}_t$ is the global model at round t , and γ_t is the magnitude of global update at round t .

In practice, the hyper-parameter γ_t is computed as follows: 1) sorting the ℓ_2 norms. The ℓ_2 norms of local model updates are sorted in ascending order during the communication round t . These norms are arranged as $a_1^t, \dots, a_i^t, a_j^t, \dots, a_{\mathcal{M}}^t$, where $\forall i < j \leq \mathcal{M}, a_i^t \leq a_j^t$. 2) detecting sudden changes. To detect sudden changes in the increase between two consecutive ℓ_2 norms, RoseAgg calculates the difference between each consecutive pair of a_i^t and a_{i+1}^t as $\Delta a_i^t = a_{i+1}^t - a_i^t$. A sudden change can be defined as a significant increase in Δa_i^t compared to the previous values. Specifically, RoseAgg sets a percentage change threshold ζ and considers any Δa_i^t that exceeds this threshold as indicating a sudden change. Formally, if $\Delta a_i^t > (1+\zeta)\Delta a_{i-1}^t$, it indicates a sudden change in the increase in ℓ_2 norms. 3) computing γ_t . If no sudden change is detected, indicating steady and continuous increases in the ℓ_2 norms, then γ_t is set as the median of all ℓ_2 norms, i.e., $\gamma_t = \text{Median}(a_1^t, a_2^t, \dots, a_{\mathcal{M}}^t)$. However, if a sudden change is identified at index i , then γ_t is set to the median of the ℓ_2 norms up to index i , i.e., $\gamma_t = \text{Median}(a_1^t, a_2^t, \dots, a_i^t)$. Since the ℓ_2 norm of each benign model update becomes smaller in later rounds, γ_t will decrease with the increase of rounds.

For ease of understanding, Figure 4 provides a toy example to illustrate the overall workflow of RoseAgg. Consider a scenario with 6 clients clustered into 2 groups using the DBSCAN algorithm, employing cosine distance as the similarity metric. Each cluster has a distance matrix, as illustrated in Figure 4. RoseAgg initially performs adaptive clustering and subsequently apply partial aggregation to local updates within

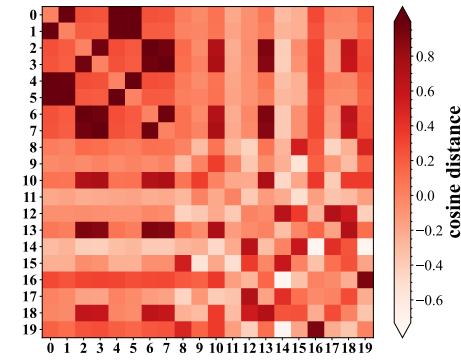


图 3. 本地更新的余弦距离矩阵。该矩阵包含从 0 到 19 的客户端 ID。客户端 0 和 4 的目标是将包含 “F” 像素模式的图像误分类为飞机。另一方面，客户端 1 和 5 的意图是将包含 “L” 像素模式的图像误识别为飞机。客户端 2 和 6 特别旨在将具有类似字母 “A” 的像素模式的图像误分类为鸟类。最后，客户端 3 和 7 专注于将背景中有垂直条纹墙壁的汽车图像误识别为鸟类。

在 $\sim$ 中, $\{v_4\}g_0^t$ 是期望的成分, $\lambda_{\max}$ 和 $\xi_{\max}$ 分别是 $\hat{\mathbf{H}}^t$ 的最大特征值和对应的特征向量。此外, 由于 $\hat{\mathbf{H}}^t \in \mathbb{R}^{K_t \times K_t}$ 是一个低维矩阵, 因此在服务器上计算 $\lambda_{\max}$ 和 $\xi_{\max}$ 的计算开销不会太大。干净的成分提取步骤在算法 2 的第 12-15 行中显示。

3) 信用评分计算: FLTrust [6] 根据 ReLU 裁剪后的余弦距离为每个客户端分配一个信任评分。然而, 这可能会将模型更新分配给与服务器维护的模型更新之间具有较大角度的 $\langle \text{样式 id}='4' \rangle 0 \langle / \text{样式} \rangle$ 。也就是说, FLTrust 将此类模型更新视为恶意更新并将其排除。然而, 此类角度偏差可能是由于数据异质性 [14], [16][24] 造成的。仅仅移除模型更新可能会导致全局模型的精度下降。因此, 需要一种替代方法。

具体来说, RoseAgg 使用每个部分聚合更新在干净成分上的投影方差作为其信用评分。形式上, 信用评分定义如下:

$$cre_i^t \leftarrow \left(\left\| \tilde{\mathbf{g}}^{t,i} \right\| \frac{\langle \tilde{\mathbf{g}}^{t,i}, \tilde{\mathbf{g}}_0^t \rangle}{\left\| \tilde{\mathbf{g}}^{t,i} \right\| \left\| \tilde{\mathbf{g}}_0^t \right\|} \right)^2, \quad i \in [1, K_t] \quad (6)$$

其中 $\langle \text{样式 id}='1' \rangle cre \langle / \text{样式} \rangle_i$ 是第 $\langle \text{样式 id}='4' \rangle i \langle / \text{样式} \rangle$ 个部分聚合更新的 $\sim \langle \text{样式 id}='7' \rangle g \langle / \text{样式} \rangle \langle \text{样式 id}='10' \rangle, \langle / \text{样式} \rangle$ 在轮次 $\langle \text{样式 id}='13' \rangle t \langle / \text{样式} \rangle$ 。以这种方式计算信用评分的原理是基于 $\sim \langle \text{样式 id}='16' \rangle g \langle / \text{样式} \rangle_0$ 是一个合理的干净成分。因此, 良性数据点的投影方差之和将大于中毒数据点的投影方差之和。结果, 在加权求和中, 良性数据点的权重将大于中毒数据点的权重, 从而减轻模型中毒攻击。

4) 更新全局模型: 全局更新是通过加权平均部分聚合的模型更新来计算的, 权重为信用分数:

$$\mathbf{g}^t \leftarrow \frac{\sum_{i=1}^{K_t} cre_i^t \tilde{\mathbf{g}}^{t,i}}{\sum_{i=1}^{K_t} cre_i^t}$$

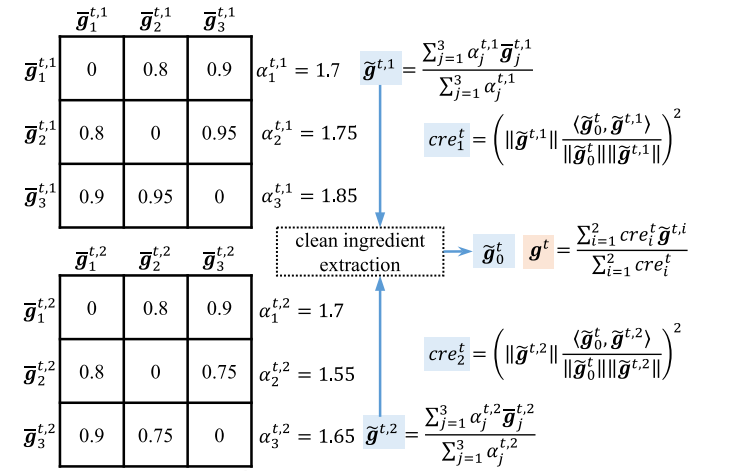


图 4. RoseAgg 方法的玩具示例。

$$\tilde{\mathbf{g}}^t \leftarrow \frac{\mathbf{g}^t}{\left\| \mathbf{g}^t \right\|} \quad (7)$$

其中 $\tilde{\mathbf{g}}^t$ 是第 t 轮的归一化全局更新。通过这种方式, RoseAgg 可以整合所有参与者的贡献, 同时更侧重于良性参与者。随后, 服务器按照以下方式更新全局模型:

$$\mathbf{w}_t \leftarrow \mathbf{w}_{t-1} - \gamma_t \tilde{\mathbf{g}}^t \quad (8)$$

其中 $\mathbf{w}_t$ 是第 t 轮的全局模型, 而 γ_t 是第 t 轮全局更新的幅度。

在实践中, 超参数 γ_t 计算如下: 1) 排序 ℓ_2 范数。在通信轮 t 中, 本地模型更新的 ℓ_2 范数按升序排序。这些范数排列为 $a_1^t, \dots, a_i^t, a_j^t, \dots, a_{\mathcal{M}}^t$, 其中 $\forall i < j \leq \mathcal{M}, a_i^t \leq a_j^t$ 。2) 检测突变。为了检测两个连续的 ℓ_2 范数之间的增量突变, RoseAgg 计算每对 a_i^t 和 a_{i+1}^t 之间的差值作为 $\Delta a_i^t = a_{i+1}^t - a_i^t$ 。突变可以定义为 Δa_i^t 与先前值相比的显著增加。具体来说, RoseAgg 设置了一个百分比变化阈值 ζ , 并将任何超过此阈值的 Δa_i^t 视为指示突变。形式上, 如果 $\Delta a_i^t > (1+\zeta)\Delta a_{i-1}^t$, 则表示 ℓ_2 范数的增量出现突变。3) 计算 γ_t 。如果没有检测到突变, 表明 ℓ_2 范数稳定且连续增加, 则 γ_t 设置为所有 ℓ_2 范数的中位数, 即 $\gamma_t = \text{Median}(a_1^t, a_2^t, \dots, a_{\mathcal{M}}^t)$ 。然而, 如果在第 i 个索引处检测到突变, 则 γ_t 设置为直到索引 i 的 ℓ_2 范数的中位数, 即 $\gamma_t = \text{Median}(a_1^t, a_2^t, \dots, a_i^t)$ 。由于每个良性模型更新的 ℓ_2 范数在后续轮次中会变小, γ_t 随着轮次的增加而减少。

为便于理解, 图 4 提供了一个玩具示例来说明 RoseAgg 的整体工作流程。考虑一个使用 DBSCAN 算法将 6 个客户端聚类成 2 个组的场景, 并采用余弦距离作为相似性度量。每个簇都有一个距离矩阵, 如图 4 所示。RoseAgg 首先执行自适应聚类, 然后对本地更新

each cluster. To exemplify this, let's take local updates within cluster 1. These updates are combined to create a partially aggregated vector $\tilde{\mathbf{g}}^{t,1}$, computed as $\tilde{\mathbf{g}}^{t,1} = \frac{\sum_{j=1}^3 \alpha_j^{t,1} \tilde{\mathbf{g}}_j^{t,1}}{\sum_{j=1}^3 \alpha_j^{t,1}}$. Here,

$\alpha_j^{t,1}$ represents the summation of the j -th row from the distance matrix of cluster 1 during the t -th round. Subsequently, a clean ingredient $\tilde{\mathbf{g}}_0^t$ is derived via the clean ingredient extraction module. Formally, let $\mathcal{H}^t = (\tilde{\mathbf{g}}^{t,1}, \tilde{\mathbf{g}}^{t,2}) \in \mathbb{R}^{d \times 2}$, where d is the dimension of the model. $\tilde{\mathbf{g}}_0^t$ corresponds precisely to the first principal component of the matrix $\hat{\mathcal{H}}^t = \mathcal{H}^t \mathcal{H}^{tT}$. Furthermore, a credit score is calculated for each partially aggregated update. Concretely, the credit score of $\tilde{\mathbf{g}}^{t,1}$ is computed as follows:

$$cre_1 = (\|\tilde{\mathbf{g}}^{t,1}\| \cos(\tilde{\mathbf{g}}_0^t, \tilde{\mathbf{g}}^{t,1}))^2 = \left(\|\tilde{\mathbf{g}}^{t,1}\| \frac{\langle \tilde{\mathbf{g}}_0^t, \tilde{\mathbf{g}}^{t,1} \rangle}{\|\tilde{\mathbf{g}}_0^t\| \|\tilde{\mathbf{g}}^{t,1}\|} \right)^2,$$

where $\langle \tilde{\mathbf{g}}_0^t, \tilde{\mathbf{g}}^{t,1} \rangle$ denotes the inner product between $\tilde{\mathbf{g}}_0^t$ and $\tilde{\mathbf{g}}^{t,1}$. Finally, the global update is determined by computing the average of the partially aggregated model updates, as follows:

$$\mathbf{g}^t = \frac{\sum_{i=1}^2 cre_i^t \tilde{\mathbf{g}}^{t,i}}{\sum_{i=1}^2 cre_i^t}.$$

VI. THEORETICAL ANALYSIS

In FL, the optimal global model $\mathbf{w}^*$ is the solution to the following optimization problem: $\mathbf{w}^* = \arg \min_{\mathbf{w} \in \Theta} F(\mathbf{w}) \triangleq \mathbb{E}_{D \sim \chi} [f(D, \mathbf{w})]$. Θ is the parameter space of the global model. $\mathcal{D}$ is the joint training dataset of the n clients. The distribution of the overall data is represented by χ . For the sake of clarity, the superscript t is omitted in this section.

Assumption 1: The expected loss function $F(\mathbf{w})$ is μ -strongly convex and differentiable over model parameter space Θ with L -Lipschitz continuous gradient. Formally, for any $\mathbf{w}, \hat{\mathbf{w}} \in \Theta$:

$$F(\hat{\mathbf{w}}) \geq F(\mathbf{w}) + \langle \nabla F(\mathbf{w}), \hat{\mathbf{w}} - \mathbf{w} \rangle + \frac{\mu}{2} \|\hat{\mathbf{w}} - \mathbf{w}\|^2,$$

$$\|\nabla F(\mathbf{w}) - \nabla F(\hat{\mathbf{w}})\| \leq L \|\mathbf{w} - \hat{\mathbf{w}}\|,$$

For the local empirical loss function $f(\mathcal{D}, \mathbf{w})$, we assume that it is probabilistically L_1 -Lipschitz. Then for any $\delta \in (0, 1)$, there exists a positive constant L_1 satisfying:

$$Pr\left\{ \sup_{\mathbf{w}, \hat{\mathbf{w}} \in \Theta; \mathbf{w} \neq \hat{\mathbf{w}}} \frac{\|\nabla f(\mathcal{D}, \mathbf{w}) - \nabla f(\mathcal{D}, \hat{\mathbf{w}})\|}{\|\mathbf{w} - \hat{\mathbf{w}}\|} \leq L_1 \right\} \geq 1 - \frac{\delta}{3}$$

Assumption 2: The gradient of the empirical loss function $\nabla f(\mathcal{D}, \mathbf{w}^)$ at the optimal global model $\mathbf{w}^*$ is bounded. Moreover, the gradient difference $h(\mathcal{D}, \mathbf{w}) = \nabla f(\mathcal{D}, \mathbf{w}) - \nabla f(\mathcal{D}, \mathbf{w}^*)$ for any $\mathbf{w} \in \Theta$ is bounded. Specifically, there exist positive constants σ_1 and κ_1 so that for any unit vector v , $\langle \nabla f(\mathcal{D}, \mathbf{w}^*), v \rangle$ is sub-exponential with σ_1 and κ_1 ; and there also exist positive constants σ_2 and κ_2 so that for any $\mathbf{w} \in \Theta$ with $\mathbf{w} \neq \hat{\mathbf{w}}$ and any unit vector v , $\langle h(\mathcal{D}, \mathbf{w}), v \rangle / \|\mathbf{w} - \mathbf{w}^*\|$ is sub-exponential with σ_2 and κ_2 . Formally, for $\forall |\xi| \leq 1/\kappa_1, \forall |\xi| \leq 1/\kappa_2$:*

$$\sup_{v \in B} \mathbb{E}[\exp(\xi \langle \nabla f(\mathcal{D}, \mathbf{w}^*), v \rangle)] \leq e^{(\sigma_1)^2 \xi^2 / 2},$$

$$\sup_{\mathbf{w} \in \Theta, v \in B} \mathbb{E}[\exp(\frac{\xi \langle h(\mathcal{D}, \mathbf{w}) - \mathbb{E}[h(\mathcal{D}, \mathbf{w})], v \rangle}{\|\mathbf{w} - \mathbf{w}^*\|})] \leq e^{(\sigma_2)^2 \xi^2 / 2},$$

where B is the unit sphere $B = \{v : \|v\| = 1\}$.

Assumption 3: γ depends on the gradient magnitude of the benign client, which is assumed to be bounded. Hence, we make the following assumptions regarding the bound of γ : $\gamma \leq \Gamma$.

Lemma 1: Assume Assumption 3 holds. The distance between $\mathbf{g} = \gamma \tilde{\mathbf{g}}$ and $\nabla F(\mathbf{w})$ is bounded as follows in each iteration:

$$\|\mathbf{g} - \nabla F(\mathbf{w})\| \leq \Gamma + G \|\nabla F(\mathbf{w})\| + (G + 1) \|\mathbf{g}_0 - \nabla F(\mathbf{w})\| \quad (9)$$

where $G = \max\{1, 2\Gamma - 1\}$. Given that $\mathbf{g}_0$ is a plausible ingredient, it can be approximately regarded as a model update resulting from the joint training of a group of benign clients. Formally, $\mathbf{g} \approx \frac{1}{n_b} \sum_{i \in \mathcal{N}_b} \frac{1}{|\mathcal{D}_i|} \sum_{X_j \in \mathcal{D}_i} \nabla f(X_j, \mathbf{w})$, $\mathcal{N}_b$ denotes the collection of benign clients, with a total count of n_b , $\mathcal{D}_i$ denotes the dataset of client i , $|\mathcal{D}_i|$ is the size of $\mathcal{D}_i$, and X_j represents the j -th data sample contained within $\mathcal{D}_i$.

Proof:

$$\begin{aligned} \|\mathbf{g} - \nabla F(\mathbf{w})\| &\leq \|\gamma \tilde{\mathbf{g}} - \mathbf{g}_0\| + \|\mathbf{g}_0 - \nabla F(\mathbf{w})\| \\ &= \|\gamma (\tilde{\mathbf{g}} - \mathbf{g}_0) - (1 - \gamma) \mathbf{g}_0\| + \|\mathbf{g}_0 - \nabla F(\mathbf{w})\| \\ &\leq \gamma(1 + \|\mathbf{g}_0\|) + |1 - \gamma| \|\mathbf{g}_0\| + \|\mathbf{g}_0 - \nabla F(\mathbf{w})\| \\ &= \gamma + (\gamma + |1 - \gamma|) \|\mathbf{g}_0\| + \|\mathbf{g}_0 - \nabla F(\mathbf{w})\| \\ &\leq \Gamma + \underbrace{\max\{1, 2\Gamma - 1\}}_G \|\mathbf{g}_0\| + \|\mathbf{g}_0 - \nabla F(\mathbf{w})\| \\ &= \Gamma + G \|\mathbf{g}_0 + \nabla F(\mathbf{w}) - \nabla F(\mathbf{w})\| + \|\mathbf{g}_0 - \nabla F(\mathbf{w})\| \\ &\leq \Gamma + G \|\nabla F(\mathbf{w})\| + (G + 1) \|\mathbf{g}_0 - \nabla F(\mathbf{w})\|. \end{aligned}$$

□

Lemma 2: Suppose Assumption 1 holds, the following would apply for any global iteration $t \geq 1$, where α as the global learning rate:

$$\begin{aligned} \|\mathbf{w}_{t-1} - \mathbf{w}^* - \alpha \nabla F(\mathbf{w}_{t-1})\|^2 &\leq (1 + \alpha^2 L^2 - \alpha \mu) \|\mathbf{w}_{t-1} - \mathbf{w}^*\|^2 \end{aligned} \quad (10)$$

Proof: Derived from $\nabla F(\mathbf{w}^*) = 0$, the following expression arises:

$$\begin{aligned} \|\mathbf{w}^{t-1} - \mathbf{w}^* - \alpha \nabla F(\mathbf{w}^{t-1})\|^2 &= \|\mathbf{w}^{t-1} - \mathbf{w}^* - \alpha (\nabla F(\mathbf{w}^{t-1}) - \nabla F(\mathbf{w}^*))\|^2 \\ &= \|\mathbf{w}^{t-1} - \mathbf{w}^*\|^2 + \alpha^2 \|\nabla F(\mathbf{w}^{t-1}) - \nabla F(\mathbf{w}^*)\|^2 \\ &\quad - 2\alpha \langle \mathbf{w}^{t-1} - \mathbf{w}^*, \nabla F(\mathbf{w}^{t-1}) - \nabla F(\mathbf{w}^*) \rangle. \end{aligned} \quad (11)$$

Based on Assumption 1, the following formulas holds:

$$\|\nabla F(\mathbf{w}^{t-1}) - \nabla F(\mathbf{w}^*)\| \leq L \|\mathbf{w}^{t-1} - \mathbf{w}^*\|, \quad (12)$$

$$\begin{aligned} F(\mathbf{w}^*) + \langle \nabla F(\mathbf{w}^*), \mathbf{w}^{t-1} - \mathbf{w}^* \rangle &\leq F(\mathbf{w}^{t-1}) - \frac{\mu}{2} \|\mathbf{w}^{t-1} - \mathbf{w}^*\|^2, \end{aligned} \quad (13)$$

$$F(\mathbf{w}^{t-1}) + \langle \nabla F(\mathbf{w}^{t-1}), \mathbf{w}^* - \mathbf{w}^{t-1} \rangle \leq F(\mathbf{w}^*), \quad (14)$$

By summing up inequalities (13) and (14), the following expression is obtained:

$$\langle \nabla F(\mathbf{w}^{t-1}) - \nabla F(\mathbf{w}^*), \mathbf{w}^* - \mathbf{w}^{t-1} \rangle \leq -\frac{\mu}{2} \|\mathbf{w}^{t-1} - \mathbf{w}^*\|^2, \quad (15)$$

每个集群。为了举例说明这一点，我们以集群 1 内的本地更新为例。这些更新被组合起来创建一个部分聚合向量 $\tilde{\mathbf{g}}^{t,1}$ ，计算方式为 $\tilde{\mathbf{g}}^{t,1} = \frac{\sum_{j=1}^3 \alpha_j^{t,1} \tilde{\mathbf{g}}_j^{t,1}}{\sum_{j=1}^3 \alpha_j^{t,1}}$ 。

这里 $\alpha_j^{t,1}$ 表示在 t 轮中集群 1 的距离矩阵的第 j 行的总和。随后，通过干净成分提取模块推导出干净的成分 $\tilde{\mathbf{g}}_0^t$ 。形式上，令 $\mathcal{H}^t = (\tilde{\mathbf{g}}^{t,1}, \tilde{\mathbf{g}}^{t,2}) \in \mathbb{R}^{d \times 2}$ ，其中 d 是模型的维度。 $\tilde{\mathbf{g}}_0^t$ 精确对应于矩阵 $\hat{\mathcal{H}}^t = \mathcal{H}^t \mathcal{H}^{tT}$ 的第一主成分。此外，为每个部分聚合更新计算信用分数。具体来说， $\tilde{\mathbf{g}}^{t,1}$ 的信用分数计算如下： $cre_1 = (\|\tilde{\mathbf{g}}^{t,1}\| \cos(\tilde{\mathbf{g}}_0^t, \tilde{\mathbf{g}}^{t,1}))^2 = \left(\|\tilde{\mathbf{g}}^{t,1}\| \frac{\langle \tilde{\mathbf{g}}_0^t, \tilde{\mathbf{g}}^{t,1} \rangle}{\|\tilde{\mathbf{g}}_0^t\| \|\tilde{\mathbf{g}}^{t,1}\|} \right)^2$ ，其中

$(\tilde{\mathbf{g}}_0^t, \tilde{\mathbf{g}}^{t,1})$ 表示 $\tilde{\mathbf{g}}_0^t$ 和 $\tilde{\mathbf{g}}^{t,1}$ 之间的内积。最后，全局更新通过计算部分聚合模型更新的平均值来确定，如下所示： $\mathbf{g}^t = \frac{\sum_{i=1}^2 cre_i^t \tilde{\mathbf{g}}^{t,i}}{\sum_{i=1}^2 cre_i^t}$ 。

VI. 理论分析

在联邦学习中，最优的全局模型 $\mathbf{w}^*$ 是以下优化问题的解： $\mathbf{w}^* = \arg \min_{\mathbf{w} \in \Theta} F(\mathbf{w}) \triangleq \mathbb{E}_{D \sim \chi} [f(D, \mathbf{w})]$ 。 Θ 是全局模型的参数空间。 $\mathcal{D}$ 是所有客户端的联合训练数据集。整体数据的分布由 χ 表示。为了清晰起见，本节省略了上标 t 。

假设 1：期望损失函数 $F(\mathbf{w})$ 在模型参数空间 Θ 上是 μ -强凸且可微的，并且梯度具有 L -Lipschitz 连续性。

形式上，对于任何 $\mathbf{w}, \hat{\mathbf{w}} \in \Theta$ ：

$$F(\hat{\mathbf{w}}) \geq F(\mathbf{w}) + \langle \nabla F(\mathbf{w}), \hat{\mathbf{w}} - \mathbf{w} \rangle + \frac{\mu}{2} \|\hat{\mathbf{w}} - \mathbf{w}\|^2,$$

$$\|\nabla F(\mathbf{w}) - \nabla F(\hat{\mathbf{w}})\| \leq L \|\mathbf{w} - \hat{\mathbf{w}}\|,$$

对于局部经验损失函数 $f(\mathcal{D}, \mathbf{w})$ ，我们假设它是概率 L_1 -Lipschitz 的。然后对于任何 $\delta \in (0, 1)$ ，存在一个正的常数 L_1 满足：

$$Pr\left\{ \sup_{\mathbf{w}, \hat{\mathbf{w}} \in \Theta; \mathbf{w} \neq \hat{\mathbf{w}}} \frac{\|\nabla f(\mathcal{D}, \mathbf{w}) - \nabla f(\mathcal{D}, \hat{\mathbf{w}})\|}{\|\mathbf{w} - \hat{\mathbf{w}}\|} \leq L_1 \right\} \geq 1 - \frac{\delta}{3}$$

假设 2：经验损失函数 $\nabla f(\mathcal{D}, \mathbf{w}^*)$ 在最优全局模型 $\mathbf{w}^*$ 处的梯度有界。此外，对于任何 $\mathbf{w} \in \Theta$ ，梯度差 $h(\mathcal{D}, \mathbf{w}) = \nabla f(\mathcal{D}, \mathbf{w}) - \nabla f(\mathcal{D}, \mathbf{w}^*)$ 有界。具体来说，存在正常数 σ_1 和 κ_1 ，使得对于任何单位向量 v ， $\langle \nabla f(\mathcal{D}, \mathbf{w}^*), v \rangle$ 关于 σ_1 和 κ_1 是次指数的；并且还存正常数 σ_2 和 κ_2 ，使得对于任何 $\mathbf{w} \in \Theta$ 满足 $\mathbf{w} \neq \hat{\mathbf{w}}$ 和任何单位向量 v ， $\langle h(\mathcal{D}, \mathbf{w}) - \mathbb{E}[h(\mathcal{D}, \mathbf{w})], v \rangle / \|\mathbf{w} - \mathbf{w}^*\|$ 关于 σ_2 和 κ_2 是次指数的。形式上，对于 $\forall |\xi| \leq 1/\kappa_1, \forall |\xi| \leq 1/\kappa_2$ ：

$$\sup_{v \in B} \mathbb{E}[\exp(\xi \langle \nabla f(\mathcal{D}, \mathbf{w}^*), v \rangle)] \leq e^{(\sigma_1)^2 \xi^2 / 2},$$

$$\sup_{\mathbf{w} \in \Theta, v \in B} \mathbb{E}[\exp(\frac{\xi \langle h(\mathcal{D}, \mathbf{w}) - \mathbb{E}[h(\mathcal{D}, \mathbf{w})], v \rangle}{\|\mathbf{w} - \mathbf{w}^*\|})] \leq e^{(\sigma_2)^2 \xi^2 / 2},$$

其中 B 是单位球 $B = \{v : \|v\| = 1\}$ 。

假设 3： γ 取决于良性客户端的梯度幅度，假设该幅度有界。因此，我们关于 γ 的界限做出以下假设： $\gamma \leq \Gamma$ 。

引理 1：假设假设 3 成立。 $\mathbf{g}$ 与 $\nabla F(\mathbf{w})$ 在每次迭代中的距离有界：

$$\|\mathbf{g} - \nabla F(\mathbf{w})\| \leq \Gamma + G \|\nabla F(\mathbf{w})\| + (G + 1) \|\mathbf{g}_0 - \nabla F(\mathbf{w})\| \quad (9)$$

其中 $G = \max\{1, 2\Gamma - 1\}$ 。考虑到 $\mathbf{g}_0$ 是一个合理的成分，可以近似视为一组良性客户端联合训练产生的模型更新。形式上， $\mathbf{g} \approx \frac{1}{n_b} \sum_{i \in \mathcal{N}_b} \frac{1}{|\mathcal{D}_i|} \sum_{X_j \in \mathcal{D}_i} \nabla f(X_j, \mathbf{w})$ ， $\mathcal{N}_b$ 表示良性客户端的集合，总数为 n_b ， $\mathcal{D}_i$ 表示客户端 i 的数据集， $|\mathcal{D}_i|$ 是 $\mathcal{D}_i$ 的大小，而 X_j 表示 $\mathcal{D}_i$ 中的第 j -个数据样本。

Proof:

$$\begin{aligned} \|\mathbf{g} - \nabla F(\mathbf{w})\| &\leq \|\gamma \tilde{\mathbf{g}} - \mathbf{g}_0\| + \|\mathbf{g}_0 - \nabla F(\mathbf{w})\| \\ &= \|\gamma (\tilde{\mathbf{g}} - \mathbf{g}_0) - (1 - \gamma) \mathbf{g}_0\| + \|\mathbf{g}_0 - \nabla F(\mathbf{w})\| \\ &\leq \gamma(1 + \|\mathbf{g}_0\|) + |1 - \gamma| \|\mathbf{g}_0\| + \|\mathbf{g}_0 - \nabla F(\mathbf{w})\| \\ &= \gamma + (\gamma + |1 - \gamma|) \|\mathbf{g}_0\| + \|\mathbf{g}_0 - \nabla F(\mathbf{w})\| \\ &\leq \Gamma + \underbrace{\max\{1, 2\Gamma - 1\}}_G \|\mathbf{g}_0\| + \|\mathbf{g}_0 - \nabla F(\mathbf{w})\| \\ &= \Gamma + G \|\mathbf{g}_0 + \nabla F(\mathbf{w}) - \nabla F(\mathbf{w})\| + \|\mathbf{g}_0 - \nabla F(\mathbf{w})\| \\ &\leq \Gamma + G \|\nabla F(\mathbf{w})\| + (G + 1) \|\mathbf{g}_0 - \nabla F(\mathbf{w})\|. \end{aligned}$$

□

引理 2：假设条件 1 成立，则对于任何全局迭代次数 $t \geq 1$ ，其中 α 作为全局学习率：

$$\begin{aligned} \|\mathbf{w}_{t-1} - \mathbf{w}^* - \alpha \nabla F(\mathbf{w}_{t-1})\|^2 &\leq (1 + \alpha^2 L^2 - \alpha \mu) \|\mathbf{w}_{t-1} - \mathbf{w}^*\|^2 \end{aligned} \quad (10)$$

证明：从 $\nabla F(\mathbf{w}^*) = 0$ 推导出，得到以下表达式：

$$\begin{aligned} \|\mathbf{w}^{t-1} - \mathbf{w}^* - \alpha \nabla F(\mathbf{w}^{t-1})\|^2 &= \|\mathbf{w}^{t-1} - \mathbf{w}^* - \alpha (\nabla F(\mathbf{w}^{t-1}) - \nabla F(\mathbf{w}^*))\|^2 \\ &= \|\mathbf{w}^{t-1} - \mathbf{w}^*\|^2 + \alpha^2 \|\nabla F(\mathbf{w}^{t-1}) - \nabla F(\mathbf{w}^*)\|^2 \\ &\quad - 2\alpha \langle \mathbf{w}^{t-1} - \mathbf{w}^*, \nabla F(\mathbf{w}^{t-1}) - \nabla F(\mathbf{w}^*) \rangle. \end{aligned} \quad (11)$$

根据假设 1，以下公式成立

$$\|\nabla F(\mathbf{w}^{t-1}) - \nabla F(\mathbf{w}^*)\| \leq L \|\mathbf{w}^{t-1} - \mathbf{w}^*\|, \quad (12)$$

$$\begin{aligned} F(\mathbf{w}^*) + \langle \nabla F(\mathbf{w}^*), \mathbf{w}^{t-1} - \mathbf{w}^* \rangle &\leq F(\mathbf{w}^{t-1}) - \frac{\mu}{2} \|\mathbf{w}^{t-1} - \mathbf{w}^*\|^2, \end{aligned} \quad (13)$$

$$F(\mathbf{w}^{t-1}) + \langle \nabla F(\mathbf{w}^{t-1}), \mathbf{w}^* - \mathbf{w}^{t-1} \rangle \leq F(\mathbf{w}^*), \quad (14)$$

通过将不等式 (13) 和 (14) 相加，得到如下表达式

$$\langle \nabla F(\mathbf{w}^{t-1}) - \nabla F(\mathbf{w}^*), \mathbf{w}^* - \mathbf{w}^{t-1} \rangle \leq -\frac{\mu}{2} \|\mathbf{w}^{t-1} - \mathbf{w}^*\|^2, \quad (15)$$

By applying inequalities (12) and (15) to (11), the following is established:

$$\|\mathbf{w}^{t-1} - \mathbf{w}^* - \alpha \nabla F(\mathbf{w}^{t-1})\|^2 \leq (1 + \alpha^2 L^2 - \alpha \mu) \|\mathbf{w}^{t-1} - \mathbf{w}^*\|^2. \quad (16)$$

□

Lemma 3: Suppose Assumption 2 holds for any benign client i , the following conditions are satisfied: $\forall \delta \in (0, 1)$ and $\mathbf{w} \in \Theta$, let $\Delta_1^i = \sqrt{2\sigma_1 \sqrt{(d \log 6 + \log(3/\delta))}/|\mathcal{D}_i|}$ and $\Delta_3^i = \sqrt{2\sigma_2 \sqrt{(d \log 6 + \log(3/\delta))}/|\mathcal{D}_i|}$. If $\Delta_1^i \leq (\sigma_1)^2/\kappa_1$, $\Delta_3^i \leq (\sigma_2)^2/\kappa_2$. Then:

$$\begin{aligned} & Pr\{\|\frac{1}{|\mathcal{D}_i|} \sum_{X_j \in \mathcal{D}_i} \nabla f(X_j, \mathbf{w}^*) - \nabla F(\mathbf{w}^*)\| \geq 2\Delta_1^i\} \\ & \leq \frac{\delta}{3n_b}, \\ & Pr\{\|\frac{1}{|\mathcal{D}_i|} \sum_{X_j \in \mathcal{D}_i} h(X_j, \mathbf{w}^*) - \mathbb{E}[h(X, \mathbf{w})]\| \geq 2\Delta_3^i \|\mathbf{w} - \mathbf{w}^*\|\} \\ & \leq \frac{\delta}{3n_b} \end{aligned} \quad (17)$$

Proof: The initial step involves establishing the validity of the first inequality presented in Lemma 3. The proof of the second inequality is omitted, as it follows a similar rationale. Let $V = \{v_1, \dots, v_{N_1}\}$ be an $\frac{1}{2}$ -cover of the unit sphere B . Following [6], [8], when $N_1 < s \cdot \log 6$, where $s = o(|\mathcal{D}_i|)$:

$$\begin{aligned} & \|\frac{1}{|\mathcal{D}_i|} \sum_{X_j \in \mathcal{D}_i} \nabla f(X_j, \mathbf{w}^*) - \nabla F(\mathbf{w}^*)\| \\ & \leq 2 \sup_{v \in V} \{ \|\frac{1}{|\mathcal{D}_i|} \sum_{X_j \in \mathcal{D}_i} \nabla f(X_j, \mathbf{w}^*) - \nabla F(\mathbf{w}^*), v\} \}. \end{aligned} \quad (18)$$

If Assumption 2 holds and $\Delta_1^i \leq (\sigma_1)^2/\kappa_1$, the following inequality is satisfied based on the concentration inequalities for sub-exponential random variables [30]:

$$\begin{aligned} & Pr\{\|\frac{1}{|\mathcal{D}_i|} \sum_{X_j \in \mathcal{D}_i} \nabla f(X_j, \mathbf{w}^*) - \nabla F(\mathbf{w}^*), v\| \geq \Delta_1^i\} \\ & \leq \exp(-|\mathcal{D}_i|(\Delta_1^i)^2/(2\sigma_1^2)). \end{aligned} \quad (19)$$

Recall that V contains at most 6^s vectors. In view of the union bound, it further yields that:

$$\begin{aligned} & Pr\{2 \sup_{v \in V} \{ \|\frac{1}{|\mathcal{D}_i|} \sum_{X_j \in \mathcal{D}_i} \nabla f(X_j, \mathbf{w}^*) - \nabla F(\mathbf{w}^*), v\} \geq 2\Delta_1^i\} \\ & \leq 6^s \exp(-|\mathcal{D}_i|(\Delta_1^i)^2/(2\sigma_1^2)) \\ & = \exp(-|\mathcal{D}_i|(\Delta_1^i)^2/(2\sigma_1^2) + s \log 6). \end{aligned} \quad (20)$$

Therefore,

$$\begin{aligned} & Pr\{\|\frac{1}{|\mathcal{D}_i|} \sum_{X_j \in \mathcal{D}_i} \nabla f(X_j, \mathbf{w}^*) - \nabla F(\mathbf{w}^*)\| \geq 2\Delta_1^i\} \\ & \leq \exp(-|\mathcal{D}_i|(\Delta_1^i)^2/(2\sigma_1^2) + s \log 6). \end{aligned} \quad (21)$$

□

Lemma 4: Suppose Assumptions 1 and 2 hold and for any benign client i , the following conditions are satisfied: let $\Theta \subset$

$\{\mathbf{w} : \|\mathbf{w} - \mathbf{w}^*\| \leq r\sqrt{s}\}$ hold for some positive parameter r . Then, for any $\delta \in (0, 1)$, if $\Delta_1^i \leq (\sigma_1)^2/\kappa_1$ and $\Delta_2^i \leq (\sigma_2)^2/\kappa_2$, the following relationship holds for any $\mathbf{w} \in \Theta$:

$$\begin{aligned} & Pr\{\|\frac{1}{|\mathcal{D}_i|} \sum_{X_j \in \mathcal{D}_i} \nabla f(X_j, \mathbf{w}) - \nabla F(\mathbf{w})\| \\ & \leq 8\Delta_2^i \|\mathbf{w} - \mathbf{w}^*\| + 4\Delta_1^i\} \geq 1 - \frac{\delta}{n_b} \end{aligned} \quad (22)$$

where $\Delta_2^i = \sigma_2 \sqrt{\frac{2}{|\mathcal{D}_i|}(K_1^i + K_2^i)}$, $K_1^i = s \log \frac{18L_2}{\sigma_2}$, $K_2^i = \frac{1}{2}s \log \frac{|\mathcal{D}_i|}{s} + \log \frac{6\sigma_2^2 r \sqrt{|\mathcal{D}_i|}}{\kappa_2 \sigma_1 \delta}$, $L_2 = \max\{L, L_1\}$.

Proof: The proof primarily relies on the ϵ -net argument [6], [8]. Additionally, the subscripts i of the variables such as τ , l^* , ϵ_l , etc., are omitted for brevity. Let

$$\tau = \frac{\kappa_2 \sigma_1}{2\sigma_2^2} \sqrt{\frac{s}{|\mathcal{D}_i|}} \quad \text{and} \quad l^* = \lceil r\sqrt{s/\tau} \rceil, \quad (23)$$

For integers $1 \leq l \leq l^*$, let $\Theta_l \subset \{\mathbf{w} : \|\mathbf{w} - \mathbf{w}^*\| \leq \tau l\}$. For a given l , let $\mathbf{w}_1, \dots, \mathbf{w}_{\epsilon_l}$ be an ϵ_l -cover of Θ_l , where ϵ_l is given by

$$\epsilon_l = \frac{\sigma_2 \tau l}{L_2} \sqrt{\frac{s}{|\mathcal{D}_i|}}, \quad (24)$$

where $L_2 = \max\{L, L_1\}$. Based on [29], it is established that $N_{\epsilon_l} \leq s \log(3\tau l/\epsilon_l)$. For any $\mathbf{w} \in \Theta_l$, there exists a j_l ($1 \leq j_l \leq N_{\epsilon_l}$) such that:

$$\|\mathbf{w} - \mathbf{w}_{j_l}\| \leq \epsilon_l. \quad (25)$$

By the triangle inequality:

$$\begin{aligned} & \|\frac{1}{|\mathcal{D}_i|} \sum_{X_j \in \mathcal{D}_i} \nabla f(X_j, \mathbf{w}) - \nabla F(\mathbf{w})\| \\ & \leq \|\nabla F(\mathbf{w}) - \nabla F(\mathbf{w}_{j_l})\| \\ & \quad + \|\frac{1}{|\mathcal{D}_i|} \sum_{X_j \in \mathcal{D}_i} \nabla f(X_j, \mathbf{w}_{j_l}) - \nabla F(\mathbf{w}_{j_l})\| \\ & \quad + \|\frac{1}{|\mathcal{D}_i|} \sum_{X_j \in \mathcal{D}_i} (\nabla f(X_j, \mathbf{w}) - \nabla f(X_j, \mathbf{w}_{j_l}))\| \end{aligned} \quad (26)$$

According to Assumption 1 and inequality (25):

$$\|\nabla F(\mathbf{w}) - \nabla F(\mathbf{w}_{j_l})\| \leq L \|\mathbf{w} - \mathbf{w}_{j_l}\| \leq L\epsilon_l. \quad (27)$$

Then, an event $\mathcal{E}_1$ is defined as follows:

$$\mathcal{E}_1 = \{ \sup_{\mathbf{w}, \hat{\mathbf{w}} \in \Theta; \mathbf{w} \neq \hat{\mathbf{w}}} \frac{\|\nabla f(\mathcal{D}, \mathbf{w}) - \nabla f(\mathcal{D}, \hat{\mathbf{w}})\|}{\|\mathbf{w} - \hat{\mathbf{w}}\|} \leq L_1 \} \quad (28)$$

According to Assumption 1, there exists a δ such that $Pr\{\mathcal{E}_1\} \geq 1 - \frac{\delta}{3n_b}$. Moreover:

$$\begin{aligned} & \sup_{\mathbf{w} \in \Theta} \|\frac{1}{|\mathcal{D}_i|} \sum_{X_j \in \mathcal{D}_i} (\nabla f(X_j, \mathbf{w}) - \nabla f(X_j, \mathbf{w}_{j_l}))\| \\ & \leq L_1 \|\mathbf{w} - \mathbf{w}_{j_l}\| \leq L_1 \epsilon_l. \end{aligned} \quad (29)$$

By the triangle inequality:

$$\|\frac{1}{|\mathcal{D}_i|} \sum_{X_j \in \mathcal{D}_i} \nabla f(X_j, \mathbf{w}_{j_l}) - \nabla F(\mathbf{w}_{j_l})\|$$

通过应用不等式 (12) 和 (15) 到 (11), 以下得到证明

$$\|\mathbf{w}^{t-1} - \mathbf{w}^* - \alpha \nabla F(\mathbf{w}^{t-1})\|^2 \leq (1 + \alpha^2 L^2 - \alpha \mu) \|\mathbf{w}^{t-1} - \mathbf{w}^*\|^2. \quad (16)$$

□

引理 3: 假设对任何良性客户端 i , 假设 2 成立, 则满足以下条件: 2, 则: $\forall \delta \in (0, 1)$ 和 $\mathbf{w} \in \Theta$, 令

$\Delta_1^i = \sqrt{2\sigma_1 \sqrt{(d \log 6 + \log(3/\delta))}/|\mathcal{D}_i|}$ 和 $\Delta_3^i = \sqrt{2\sigma_2 \sqrt{(d \log 6 + \log(3/\delta))}/|\mathcal{D}_i|}$. 如果 $\Delta_1^i \leq (\sigma_1)^2/\kappa_1$, 则:

$$\begin{aligned} & Pr\{\|\frac{1}{|\mathcal{D}_i|} \sum_{X_j \in \mathcal{D}_i} \nabla f(X_j, \mathbf{w}^*) - \nabla F(\mathbf{w}^*)\| \geq 2\Delta_1^i\} \\ & \leq \frac{\delta}{3n_b}, \\ & Pr\{\|\frac{1}{|\mathcal{D}_i|} \sum_{X_j \in \mathcal{D}_i} h(X_j, \mathbf{w}^*) - \mathbb{E}[h(X, \mathbf{w})]\| \geq 2\Delta_3^i \|\mathbf{w} - \mathbf{w}^*\|\} \\ & \leq \frac{\delta}{3n_b} \end{aligned} \quad (17)$$

Proof: 证明基于 ϵ -网论 [6], [8]. 此外, 为了简洁起见, 省略了变量 (如 τ 、 i^* 、 ϵ_l 等的) 下标. 令

$$\begin{aligned} & \|\frac{1}{|\mathcal{D}_i|} \sum_{X_j \in \mathcal{D}_i} \nabla f(X_j, \mathbf{w}^*) - \nabla F(\mathbf{w}^*)\| \\ & \leq 2 \sup_{v \in V} \{ \|\frac{1}{|\mathcal{D}_i|} \sum_{X_j \in \mathcal{D}_i} \nabla f(X_j, \mathbf{w}^*) - \nabla F(\mathbf{w}^*), v\} \}. \end{aligned} \quad (18)$$

如果假设 2 成立, 并且 $\Delta_1^i \leq (\sigma_1)^2/\kappa_1$, 则根据次指数随机变量的集中不等式, 以下不等式成立 [30]:

$$\begin{aligned} & Pr\{\|\frac{1}{|\mathcal{D}_i|} \sum_{X_j \in \mathcal{D}_i} \nabla f(X_j, \mathbf{w}^*) - \nabla F(\mathbf{w}^*), v\| \geq \Delta_1^i\} \\ & \leq \exp(-|\mathcal{D}_i|(\Delta_1^i)^2/(2\sigma_1^2)). \end{aligned} \quad (19)$$

回想 V 最多包含 6^s 个向量. 根据并集界限, 它进一步得出:

$$\begin{aligned} & Pr\{2 \sup_{v \in V} \{ \|\frac{1}{|\mathcal{D}_i|} \sum_{X_j \in \mathcal{D}_i} \nabla f(X_j, \mathbf{w}^*) - \nabla F(\mathbf{w}^*), v\} \geq 2\Delta_1^i\} \\ & \leq 6^s \exp(-|\mathcal{D}_i|(\Delta_1^i)^2/(2\sigma_1^2)) \\ & = \exp(-|\mathcal{D}_i|(\Delta_1^i)^2/(2\sigma_1^2) + s \log 6). \end{aligned} \quad (20)$$

因此,

$$\begin{aligned} & Pr\{\|\frac{1}{|\mathcal{D}_i|} \sum_{X_j \in \mathcal{D}_i} \nabla f(X_j, \mathbf{w}^*) - \nabla F(\mathbf{w}^*)\| \geq 2\Delta_1^i\} \\ & \leq \exp(-|\mathcal{D}_i|(\Delta_1^i)^2/(2\sigma_1^2) + s \log 6). \end{aligned} \quad (21)$$

□

引理 4: 假设假设 1 和 2 成立, 并且对于任何良性客户端 i , 以下条件得到满足: 令 $\Theta \subset$

$\{\mathbf{w} : \|\mathbf{w} - \mathbf{w}^*\| \leq r\sqrt{s}\}$ 对于某个正参数 r 成立. 然后, 对于任何 $\delta \in (0, 1)$, 如果 $\Delta_1^i \leq (\sigma_1)^2/\kappa_1$ 和 $\Delta_2^i \leq (\sigma_2)^2/\kappa_2$, 则对于任何 $\mathbf{w} \in \Theta$, 以下关系成立:

$$\begin{aligned} & Pr\{\|\frac{1}{|\mathcal{D}_i|} \sum_{X_j \in \mathcal{D}_i} \nabla f(X_j, \mathbf{w}) - \nabla F(\mathbf{w})\| \\ & \leq 8\Delta_2^i \|\mathbf{w} - \mathbf{w}^*\| + 4\Delta_1^i\} \geq 1 - \frac{\delta}{n_b} \end{aligned} \quad (22)$$

where $\Delta_2^i = \sigma_2 \sqrt{\frac{2}{|\mathcal{D}_i|}(K_1^i + K_2^i)}$, $K_1^i = s \log \frac{18L_2}{\sigma_2}$, $K_2^i = \frac{1}{2}s \log \frac{|\mathcal{D}_i|}{s} + \log \frac{6\sigma_2^2 r \sqrt{|\mathcal{D}_i|}}{\kappa_2 \sigma_1 \delta}$, $L_2 = \max\{L, L_1\}$.

证明: 证明基于 ϵ -网论 [6], [8]. 此外, 为了简洁起见, 省略了变量 (如 τ 、 i^* 、 ϵ_l 等的) 下标. 令

$$\tau = \frac{\kappa_2 \sigma_1}{2\sigma_2^2} \sqrt{\frac{s}{|\mathcal{D}_i|}} \quad \text{and} \quad l^* = \lceil r\sqrt{s/\tau} \rceil, \quad (23)$$

对于整数 $1 \leq l \leq l^*$, 令 $\Theta_l \subset \{\mathbf{w} : \|\mathbf{w} - \mathbf{w}^*\| \leq \tau l\}$. 对于给定的 l , 令 $\mathbf{w}_1, \dots, \mathbf{w}_{\epsilon_l}$ 是 Θ_l 的一个 ϵ_l -覆盖, 其中 ϵ_l 由

$$\epsilon_l = \frac{\sigma_2 \tau l}{L_2} \sqrt{\frac{s}{|\mathcal{D}_i|}}, \quad (24)$$

其中 $L_2 = \max\{L, L_1\}$. 基于 [29], 可以证明 $N_{\epsilon_l} \leq s \log(3\tau l/\epsilon_l)$. 对于任何 $\mathbf{w} \in \Theta_l$, 存在一个 j_l ($1 \leq j_l \leq N_{\epsilon_l}$) 使得:

$$\|\mathbf{w} - \mathbf{w}_{j_l}\| \leq \epsilon_l. \quad (25)$$

根据三角不等式:

$$\begin{aligned} & \|\frac{1}{|\mathcal{D}_i|} \sum_{X_j \in \mathcal{D}_i} \nabla f(X_j, \mathbf{w}) - \nabla F(\mathbf{w})\| \\ & \leq \|\nabla F(\mathbf{w}) - \nabla F(\mathbf{w}_{j_l})\| \\ & \quad + \|\frac{1}{|\mathcal{D}_i|} \sum_{X_j \in \mathcal{D}_i} \nabla f(X_j, \mathbf{w}_{j_l}) - \nabla F(\mathbf{w}_{j_l})\| \\ & \quad + \|\frac{1}{|\mathcal{D}_i|} \sum_{X_j \in \mathcal{D}_i} (\nabla f(X_j, \mathbf{w}) - \nabla f(X_j, \mathbf{w}_{j_l}))\| \end{aligned} \quad (26)$$

根据假设 1 和不等式 (25):

$$\|\nabla F(\mathbf{w}) - \nabla F(\mathbf{w}_{j_l})\| \leq L \|\mathbf{w} - \mathbf{w}_{j_l}\| \leq L\epsilon_l. \quad (27)$$

然后, 定义事件 $\mathcal{E}_1$ 如下:

$$\mathcal{E}_1 = \{ \sup_{\mathbf{w}, \hat{\mathbf{w}} \in \Theta; \mathbf{w} \neq \hat{\mathbf{w}}} \frac{\|\nabla f(\mathcal{D}, \mathbf{w}) - \nabla f(\mathcal{D}, \hat{\mathbf{w}})\|}{\|\mathbf{w} - \hat{\mathbf{w}}\|} \leq L_1 \} \quad (28)$$

根据假设 1, 存在一个 δ 使得 $Pr\{\mathcal{E}_1\} \geq 1 - \frac{\delta}{3n_b}$. 此外:

$$\begin{aligned} & \sup_{\mathbf{w} \in \Theta} \|\frac{1}{|\mathcal{D}_i|} \sum_{X_j \in \mathcal{D}_i} (\nabla f(X_j, \mathbf{w}) - \nabla f(X_j, \mathbf{w}_{j_l}))\| \\ & \leq L_1 \|\mathbf{w} - \mathbf{w}_{j_l}\| \leq L_1 \epsilon_l. \end{aligned} \quad (29)$$

根据三角不等式:

$$\|\frac{1}{|\mathcal{D}_i|} \sum_{X_j \in \mathcal{D}_i} \nabla f(X_j, \mathbf{w}_{j_l}) - \nabla F(\mathbf{w}_{j_l})\|$$

$$\begin{aligned}
&\leq \left\| \frac{1}{|\mathcal{D}_i|} \sum_{X_j \in \mathcal{D}_i} \nabla f(X_j, \mathbf{w}^*) - \nabla F(\mathbf{w}^*) \right\| \\
&\quad + \left\| \frac{1}{|\mathcal{D}_i|} \sum_{X_j \in \mathcal{D}_i} (\nabla f(X_j, \mathbf{w}_{j_l}) - \nabla f(X_j, \mathbf{w}^*)) \right. \\
&\quad \left. - (\nabla F(\mathbf{w}_{j_l}) - \nabla F(\mathbf{w}^*)) \right\| \\
&\leq \left\| \frac{1}{|\mathcal{D}_i|} \sum_{X_j \in \mathcal{D}_i} \nabla f(X_j, \mathbf{w}^*) - \nabla F(\mathbf{w}^*) \right\| \\
&\quad + \left\| \frac{1}{|\mathcal{D}_i|} \sum_{X_j \in \mathcal{D}_i} h(X_j, \mathbf{w}_{j_l}) - \mathbb{E}[h(X, \mathbf{w}_{j_l})] \right\|, \quad (30)
\end{aligned}$$

where $\mathbb{E}[h(X, \mathbf{w})] = \nabla F(\mathbf{w}) - \nabla F(\mathbf{w}^*)$.

Then events $\mathcal{E}_2$ and $\mathcal{E}_3(l)$ are defined as:

$$\begin{aligned}
\mathcal{E}_2 &= \{ \left\| \frac{1}{|\mathcal{D}_i|} \sum_{X_j \in \mathcal{D}_i} \nabla f(X_j, \mathbf{w}^*) - \nabla F(\mathbf{w}^*) \right\| \leq 2\Delta_1^i \}, \\
\mathcal{E}_3(l) &= \{ \sup_{1 \leq k \leq N_\epsilon} \left\| \frac{1}{|\mathcal{D}_i|} \sum_{X_j \in \mathcal{D}_i} h(X_j, \mathbf{w}_k) - \mathbb{E}[h(X, \mathbf{w}_k)] \right\| \leq 2\Delta_2^i \tau l \}. \quad (31)
\end{aligned}$$

Since $\Delta_1^i \leq \sigma_1^2/\kappa_1$, it follows from Lemma 3 that $Pr\{\mathcal{E}_2\} \geq 1 - \delta/(3n_b)$. Similarly, based on Lemma 3 and [8], $Pr\{\mathcal{E}_3(l)\} \geq 1 - \delta/(3n_b l^*)$. Therefore, on event $\mathcal{E}_1 \cap \mathcal{E}_2 \cap \mathcal{E}_3(l)$:

$$\begin{aligned}
&\sup_{\mathbf{w} \in \Theta_l} \left\| \frac{1}{|\mathcal{D}_i|} \sum_{X_j \in \mathcal{D}_i} \nabla f(X_j, \mathbf{w}) - \nabla F(\mathbf{w}) \right\| \\
&\leq L\epsilon_l + L_1\epsilon_l + 2\Delta_1^i + 2\Delta_2^i \tau l \\
&\stackrel{(a)}{\leq} 2L_2\epsilon_l + 2\Delta_1^i + 2\Delta_2^i \tau l \stackrel{(b)}{\leq} 4\Delta_2^i \tau l + 2\Delta_1^i, \quad (32)
\end{aligned}$$

where (a) holds for $L + L_1 \leq 2L_2$ and (b) is due to $L_2\epsilon_l \leq \Delta_2^i \tau l$. Thus, according to the union bound, the probability is at least $1 - \delta/n_b$ that event $\mathcal{E}_1 \cap \mathcal{E}_2 \cap (\bigcap_{l=1}^{l^*} \mathcal{E}_3(l))$ holds. Then

on event $\mathcal{E}_1 \cap \mathcal{E}_2 \cap (\bigcap_{l=1}^{l^*} \mathcal{E}_3(l))$, for any $\mathbf{w} \in \Theta_{l^*}$, there exists an $1 \leq l \leq l^*$ such that $(l-1)\tau \leq \|\mathbf{w} - \mathbf{w}^*\| \leq l\tau$ holds. If $l = 1$, then:

$$\begin{aligned}
\left\| \frac{1}{|\mathcal{D}_i|} \sum_{X_j \in \mathcal{D}_i} \nabla f(X_j, \mathbf{w}) - \nabla F(\mathbf{w}) \right\| &\leq 4\Delta_2^i \tau + 2\Delta_1^i \\
&\stackrel{(a)}{\leq} 4\Delta_1^i, \quad (33)
\end{aligned}$$

where (a) holds for $\Delta_2^i \leq \sigma_2^2/\kappa_2$ and $\Delta_1^i \geq \sigma_1\sqrt{s/|\mathcal{D}_i|}$. If $l \geq 2$ then $2(l-1) \geq l$ and the following inequality holds:

$$\begin{aligned}
\left\| \frac{1}{|\mathcal{D}_i|} \sum_{X_j \in \mathcal{D}_i} \nabla f(X_j, \mathbf{w}) - \nabla F(\mathbf{w}) \right\| &\leq 8\Delta_2^i \|\mathbf{w} - \mathbf{w}^*\| \\
&\quad + 2\Delta_1^i. \quad (34)
\end{aligned}$$

By combining $l = 1$ and $l \geq 2$:

$$\begin{aligned}
\left\| \frac{1}{|\mathcal{D}_i|} \sum_{X_j \in \mathcal{D}_i} \nabla f(X_j, \mathbf{w}) - \nabla F(\mathbf{w}) \right\| &\leq 8\Delta_2^i \|\mathbf{w} - \mathbf{w}^*\| \\
&\quad + 4\Delta_1^i. \quad (35)
\end{aligned}$$

Lemma 5: Suppose Assumptions 1 and 2 hold and based on Lemma 4, it can be deduced that, for any $\delta \in (0, 1)$, if $\Delta_1^i \leq (\sigma_1)^2/\kappa_1$ and $\Delta_2^i \leq (\sigma_2)^2/\kappa_2$, the following holds for any $\mathbf{w} \in \Theta \subset \{\mathbf{w} : \|\mathbf{w} - \mathbf{w}^\| \leq r\sqrt{s}\}$:*

$$\begin{aligned}
&Pr\left\{ \left\| \frac{1}{n_b} \sum_{i \in \mathcal{N}_b} \frac{1}{|\mathcal{D}_i|} \sum_{X_j \in \mathcal{D}_i} \nabla f(X_j, \mathbf{w}) - \nabla F(\mathbf{w}) \right\| \right. \\
&\quad \left. \leq 8\Delta_2 \|\mathbf{w} - \mathbf{w}^*\| + 4\Delta_1 \right\} \\
&\geq 1 - \delta \quad (36)
\end{aligned}$$

where $\Delta_2 = \frac{1}{n_b} \sum_{i \in \mathcal{N}_b} \Delta_2^i$, $\Delta_1 = \frac{1}{n_b} \sum_{i \in \mathcal{N}_b} \Delta_1^i$ and $\mathcal{N}_b$ is the set of benign clients whose number is n_b .

Proof:

$$\begin{aligned}
&Pr\left\{ \left\| \frac{1}{n_b} \sum_{i \in \mathcal{N}_b} \frac{1}{|\mathcal{D}_i|} \sum_{X_j \in \mathcal{D}_i} \nabla f(X_j, \mathbf{w}) - \nabla F(\mathbf{w}) \right\| \right. \\
&\quad \left. > 8\Delta_2 \|\mathbf{w} - \mathbf{w}^*\| + 4\Delta_1 \right\} \\
&\leq Pr\left\{ \frac{1}{n_b} \sum_{i \in \mathcal{N}_b} \left\| \frac{1}{|\mathcal{D}_i|} \sum_{X_j \in \mathcal{D}_i} \nabla f(X_j, \mathbf{w}) - \nabla F(\mathbf{w}) \right\| \right. \\
&\quad \left. > 8\Delta_2 \|\mathbf{w} - \mathbf{w}^*\| + 4\Delta_1 \right\} \\
&\leq Pr\left\{ \bigcup_{i \in \mathcal{N}_b} \left\| \frac{1}{|\mathcal{D}_i|} \sum_{X_j \in \mathcal{D}_i} \nabla f(X_j, \mathbf{w}) - \nabla F(\mathbf{w}) \right\| \right. \\
&\quad \left. > 8\Delta_2^i \|\mathbf{w} - \mathbf{w}^*\| + 4\Delta_1^i \right\} \\
&\leq \sum_{i \in \mathcal{N}_b} \frac{\delta}{n_b} = \delta \quad (37)
\end{aligned}$$

□

Theorem 1: Suppose Assumption 1, 2, and 3 hold. The difference between the global model learned by RoseAgg and the optimal global model $\mathbf{w}^$ under no attacks is bounded. Formally, the probability of the following inequality being satisfied is at least $1 - \delta$:*

$$\begin{aligned}
\|\mathbf{w}_t - \mathbf{w}^*\| &\leq (1 - \rho)^t \|\mathbf{w}^0 - \mathbf{w}^*\| + (4\Delta_1(G + 1) + \Gamma)/\rho \\
\rho &= 1 - (\sqrt{(1 + \alpha^2 L^2 - \alpha\mu)} + 8\Delta_2(G + 1) \\
&\quad + L(G + 1 - \alpha)).
\end{aligned}$$

Proof: Using the lemmas above, the Theorem 1 can be proofed. For the t -th global iteration:

$$\begin{aligned}
&\|\mathbf{w}_t - \mathbf{w}^*\| \\
&= \|\mathbf{w}_{t-1} - \mathbf{g}^{t-1} - \mathbf{w}^*\| \\
&= \|\mathbf{w}_{t-1} - \alpha \nabla F(\mathbf{w}_{t-1}) - \mathbf{w}^* + \alpha \nabla F(\mathbf{w}_{t-1}) - \mathbf{g}^{t-1}\| \\
&\leq \|\mathbf{w}_{t-1} - \mathbf{w}^* - \alpha \nabla F(\mathbf{w}_{t-1})\| \\
&\quad + \|\nabla F(\mathbf{w}_{t-1}) - \mathbf{g}^{t-1} + (\alpha - 1) \nabla F(\mathbf{w}_{t-1})\| \\
&\leq \|\mathbf{w}_{t-1} - \mathbf{w}^* - \alpha \nabla F(\mathbf{w}_{t-1})\| \\
&\quad + \|\nabla F(\mathbf{w}_{t-1}) - \mathbf{g}^{t-1}\| + \|(1 - \alpha) \nabla F(\mathbf{w}_{t-1})\| \\
&\leq \|\mathbf{w}_{t-1} - \mathbf{w}^* - \alpha \nabla F(\mathbf{w}_{t-1})\| + (G + 1) \|\mathbf{g}_0^{t-1} - \nabla F(\mathbf{w}_{t-1})\| \\
&\quad + (G + 1 - \alpha) \|\nabla F(\mathbf{w}_{t-1})\| + \Gamma \\
&\leq \sqrt{(1 + \alpha^2 L^2 - \alpha\mu)} \|\mathbf{w}_{t-1} - \mathbf{w}^*\| \\
&\quad + (G + 1)(8\Delta_2 \|\mathbf{w}_{t-1} - \mathbf{w}^*\| + 4\Delta_1)
\end{aligned}$$

□

$$\begin{aligned}
&\leq \left\| \frac{1}{|\mathcal{D}_i|} \sum_{X_j \in \mathcal{D}_i} \nabla f(X_j, \mathbf{w}^*) - \nabla F(\mathbf{w}^*) \right\| \\
&\quad + \left\| \frac{1}{|\mathcal{D}_i|} \sum_{X_j \in \mathcal{D}_i} (\nabla f(X_j, \mathbf{w}_{j_l}) - \nabla f(X_j, \mathbf{w}^*)) \right. \\
&\quad \left. - (\nabla F(\mathbf{w}_{j_l}) - \nabla F(\mathbf{w}^*)) \right\| \\
&\leq \left\| \frac{1}{|\mathcal{D}_i|} \sum_{X_j \in \mathcal{D}_i} \nabla f(X_j, \mathbf{w}^*) - \nabla F(\mathbf{w}^*) \right\| \\
&\quad + \left\| \frac{1}{|\mathcal{D}_i|} \sum_{X_j \in \mathcal{D}_i} h(X_j, \mathbf{w}_{j_l}) - \mathbb{E}[h(X, \mathbf{w}_{j_l})] \right\|, \quad (30)
\end{aligned}$$

where $\mathbb{E}[h(X, \mathbf{w})] = \nabla F(\mathbf{w}) - \nabla F(\mathbf{w}^*)$.

Then events $\mathcal{E}_2$ and $\mathcal{E}_3(l)$ are defined as:

$$\begin{aligned}
\mathcal{E}_2 &= \{ \left\| \frac{1}{|\mathcal{D}_i|} \sum_{X_j \in \mathcal{D}_i} \nabla f(X_j, \mathbf{w}^*) - \nabla F(\mathbf{w}^*) \right\| \leq 2\Delta_1^i \}, \\
\mathcal{E}_3(l) &= \{ \sup_{1 \leq k \leq N_\epsilon} \left\| \frac{1}{|\mathcal{D}_i|} \sum_{X_j \in \mathcal{D}_i} h(X_j, \mathbf{w}_k) - \mathbb{E}[h(X, \mathbf{w}_k)] \right\| \leq 2\Delta_2^i \tau l \}. \quad (31)
\end{aligned}$$

Since $\Delta_1^i \leq \sigma_1^2/\kappa_1$, it follows from Lemma 3 that $Pr\{\mathcal{E}_2\} \geq 1 - \delta/(3n_b)$. Similarly, based on Lemma 3 and [8], $Pr\{\mathcal{E}_3(l)\} \geq 1 - \delta/(3n_b l^*)$. Therefore, on event $\mathcal{E}_1 \cap \mathcal{E}_2 \cap \mathcal{E}_3(l)$:

$$\begin{aligned}
&\sup_{\mathbf{w} \in \Theta_l} \left\| \frac{1}{|\mathcal{D}_i|} \sum_{X_j \in \mathcal{D}_i} \nabla f(X_j, \mathbf{w}) - \nabla F(\mathbf{w}) \right\| \\
&\leq L\epsilon_l + L_1\epsilon_l + 2\Delta_1^i + 2\Delta_2^i \tau l \\
&\stackrel{(a)}{\leq} 2L_2\epsilon_l + 2\Delta_1^i + 2\Delta_2^i \tau l \stackrel{(b)}{\leq} 4\Delta_2^i \tau l + 2\Delta_1^i, \quad (32)
\end{aligned}$$

where (a) holds for $L + L_1 \leq 2L_2$ and (b) is due to $L_2\epsilon_l \leq \Delta_2^i \tau l$. Thus, according to the union bound, the probability is at least $1 - \delta/n_b$ that event $\mathcal{E}_1 \cap \mathcal{E}_2 \cap (\bigcap_{l=1}^{l^*} \mathcal{E}_3(l))$ holds. Then

on event $\mathcal{E}_1 \cap \mathcal{E}_2 \cap (\bigcap_{l=1}^{l^*} \mathcal{E}_3(l))$, for any $\mathbf{w} \in \Theta_{l^*}$, there exists an $1 \leq l \leq l^*$ such that $(l-1)\tau \leq \|\mathbf{w} - \mathbf{w}^*\| \leq l\tau$ holds. If $l = 1$, then:

$$\begin{aligned}
\left\| \frac{1}{|\mathcal{D}_i|} \sum_{X_j \in \mathcal{D}_i} \nabla f(X_j, \mathbf{w}) - \nabla F(\mathbf{w}) \right\| &\leq 4\Delta_2^i \tau + 2\Delta_1^i \\
&\stackrel{(a)}{\leq} 4\Delta_1^i, \quad (33)
\end{aligned}$$

在 $\Delta_2^i \leq \sigma_2^2/\kappa_2$ 和 $\Delta_1^i \geq \sigma_1\sqrt{s/|\mathcal{D}_i|}$ 满足 (a) 的条件下 $s/|\mathcal{D}_i|$ 。如果 $l \geq 2$, 则 $2(l-1) \geq l$, 并且满足以下不等式:

$$\begin{aligned}
\left\| \frac{1}{|\mathcal{D}_i|} \sum_{X_j \in \mathcal{D}_i} \nabla f(X_j, \mathbf{w}) - \nabla F(\mathbf{w}) \right\| &\leq 8\Delta_2^i \|\mathbf{w} - \mathbf{w}^*\| \\
&\quad + 2\Delta_1^i. \quad (34)
\end{aligned}$$

By combining $l = 1$ and $l \geq 2$:

$$\begin{aligned}
\left\| \frac{1}{|\mathcal{D}_i|} \sum_{X_j \in \mathcal{D}_i} \nabla f(X_j, \mathbf{w}) - \nabla F(\mathbf{w}) \right\| &\leq 8\Delta_2^i \|\mathbf{w} - \mathbf{w}^*\| \\
&\quad + 4\Delta_1^i. \quad (35)
\end{aligned}$$

引理 5: 假设假设 1 和 2 成立, 并且基于引理 4, 可以推导出, 对于任何 $\delta \in (0, 1)$, 如果 $\Delta_1^i \leq (\sigma_1)^2/\kappa_1$ 和 $\Delta_2^i \leq (\sigma_2)^2/\kappa_2$, 则对于任何 $\mathbf{w} \in \Theta \subset \{\mathbf{w} : \|\mathbf{w} - \mathbf{w}^*\| \leq r\sqrt{s}\}$:

$$\begin{aligned}
&Pr\left\{ \left\| \frac{1}{n_b} \sum_{i \in \mathcal{N}_b} \frac{1}{|\mathcal{D}_i|} \sum_{X_j \in \mathcal{D}_i} \nabla f(X_j, \mathbf{w}) - \nabla F(\mathbf{w}) \right\| \right. \\
&\quad \left. \leq 8\Delta_2 \|\mathbf{w} - \mathbf{w}^*\| + 4\Delta_1 \right\} \\
&\geq 1 - \delta \quad (36)
\end{aligned}$$

其中 $\Delta_2 = \frac{1}{n_b} \sum_{i \in \mathcal{N}_b} \Delta_2^i$, $\Delta_1 = \frac{1}{n_b} \sum_{i \in \mathcal{N}_b} \Delta_1^i$ 和 $\mathcal{N}_b$ 是良性客户端的集合, 其数量为 n_b 。证明:

$$\begin{aligned}
&Pr\left\{ \left\| \frac{1}{n_b} \sum_{i \in \mathcal{N}_b} \frac{1}{|\mathcal{D}_i|} \sum_{X_j \in \mathcal{D}_i} \nabla f(X_j, \mathbf{w}) - \nabla F(\mathbf{w}) \right\| \right. \\
&\quad \left. > 8\Delta_2 \|\mathbf{w} - \mathbf{w}^*\| + 4\Delta_1 \right\} \\
&\leq Pr\left\{ \frac{1}{n_b} \sum_{i \in \mathcal{N}_b} \left\| \frac{1}{|\mathcal{D}_i|} \sum_{X_j \in \mathcal{D}_i} \nabla f(X_j, \mathbf{w}) - \nabla F(\mathbf{w}) \right\| \right. \\
&\quad \left. > 8\Delta_2 \|\mathbf{w} - \mathbf{w}^*\| + 4\Delta_1 \right\} \\
&\leq Pr\left\{ \bigcup_{i \in \mathcal{N}_b} \left\| \frac{1}{|\mathcal{D}_i|} \sum_{X_j \in \mathcal{D}_i} \nabla f(X_j, \mathbf{w}) - \nabla F(\mathbf{w}) \right\| \right. \\
&\quad \left. > 8\Delta_2^i \|\mathbf{w} - \mathbf{w}^*\| + 4\Delta_1^i \right\} \\
&\leq \sum_{i \in \mathcal{N}_b} \frac{\delta}{n_b} = \delta \quad (37)
\end{aligned}$$

□

定理 1: 假设假设 1、2、3 成立。RoseAgg 学习到的全局模型与无攻击情况下的最优全局模型 $\mathbf{w}^*$ 之间的差异是有界的。形式上, 满足以下不等式的概率至少为 $1 - \delta$:

$$\begin{aligned}
\|\mathbf{w}_t - \mathbf{w}^*\| &\leq (1 - \rho)^t \|\mathbf{w}^0 - \mathbf{w}^*\| + (4\Delta_1(G + 1) + \Gamma)/\rho \\
\rho &= 1 - (\sqrt{(1 + \alpha^2 L^2 - \alpha\mu)} + 8\Delta_2(G + 1) \\
&\quad + L(G + 1 - \alpha)).
\end{aligned}$$

证明: 利用上述引理, 可以证明定理 1。对于 t -次全局迭代:

$$\begin{aligned}
&\|\mathbf{w}_t - \mathbf{w}^*\| \\
&= \|\mathbf{w}_{t-1} - \mathbf{g}^{t-1} - \mathbf{w}^*\| \\
&= \|\mathbf{w}_{t-1} - \alpha \nabla F(\mathbf{w}_{t-1}) - \mathbf{w}^* + \alpha \nabla F(\mathbf{w}_{t-1}) - \mathbf{g}^{t-1}\| \\
&\leq \|\mathbf{w}_{t-1} - \mathbf{w}^* - \alpha \nabla F(\mathbf{w}_{t-1})\| \\
&\quad + \|\nabla F(\mathbf{w}_{t-1}) - \mathbf{g}^{t-1} + (\alpha - 1) \nabla F(\mathbf{w}_{t-1})\| \\
&\leq \|\mathbf{w}_{t-1} - \mathbf{w}^* - \alpha \nabla F(\mathbf{w}_{t-1})\| \\
&\quad + \|\nabla F(\mathbf{w}_{t-1}) - \mathbf{g}^{t-1}\| + \|(1 - \alpha) \nabla F(\mathbf{w}_{t-1})\| \\
&\leq \|\mathbf{w}_{t-1} - \mathbf{w}^* - \alpha \nabla F(\mathbf{w}_{t-1})\| + (G + 1) \|\mathbf{g}_0^{t-1} - \nabla F(\mathbf{w}_{t-1})\| \\
&\quad + (G + 1 - \alpha) \|\nabla F(\mathbf{w}_{t-1})\| + \Gamma \\
&\leq \sqrt{(1 + \alpha^2 L^2 - \alpha\mu)} \|\mathbf{w}_{t-1} - \mathbf{w}^*\| \\
&\quad + (G + 1)(8\Delta_2 \|\mathbf{w}_{t-1} - \mathbf{w}^*\| + 4\Delta_1)
\end{aligned}$$

$$\begin{aligned}
& + (G + 1 - \alpha) \|\nabla F(\mathbf{w}_{t-1}) - \nabla F(\mathbf{w}^*)\| + \Gamma \\
& \leq \sqrt{(1 + \alpha^2 L^2 - \alpha\mu) \|\mathbf{w}_{t-1} - \mathbf{w}^*\|} \\
& \quad + (G + 1)(8\Delta_2 \|\mathbf{w}_{t-1} - \mathbf{w}^*\| \\
& \quad + 4\Delta_1) + L(G + 1 - \alpha) \|\mathbf{w}_{t-1} - \mathbf{w}^*\| + \Gamma \\
& \leq (\sqrt{(1 + \alpha^2 L^2 - \alpha\mu) + 8\Delta_2(G + 1) + L(G + 1 - \alpha)}) \\
& \quad \|\mathbf{w}_{t-1} - \mathbf{w}^*\| + 4\Delta_1(G + 1) + \Gamma \quad (38)
\end{aligned}$$

Recursively applying the inequality (38) for T global iterations, the following conclusions can be drawn:

$$\|\mathbf{w}_T - \mathbf{w}^*\| \leq (1 - \rho)^T \|\mathbf{w}_0 - \mathbf{w}^*\| + (4\Delta_1(G + 1) + \Gamma)/\rho, \quad (39)$$

□

When $|1 - \rho| < 1$, the following conclusion can be drawn: $\lim_{T \rightarrow \infty} \|\mathbf{w}_T - \mathbf{w}^*\| \leq (4\Delta_1(G + 1) + \Gamma)/\rho$. After undergoing several training rounds, the discrepancy between the global model obtained through RoseAgg and the non-attacked global model acquired through FedAVG will be confined to a relatively narrow range. In other words, RoseAgg is robust to targeted collusion attacks in FL.

VII. EXPERIMENTAL RESULTS

This section presents a thorough evaluation of RoseAgg on four benchmark datasets, aiming to illustrate its effectiveness. The results substantiate its success in defending against state-of-the-art targeted model poisoning attacks [2], [3], [19], [31], [35], [39] (as described in Section II-B) in different settings. Meanwhile, a comprehensive comparison is conducted between RoseAgg and state-of-the-art defenses [6], [11], [22], [23] (as described in Section II-C) against these attacks.

A. Experimental Setup

1) *Datasets*: Four benchmark image classification datasets are employed in our evaluation to illustrate the effectiveness of RoseAgg. Following the previous work [35], which utilizes a Dirichlet distribution with a hyperparameter of 0.5 to partition the training data. Specifically, the training data is partitioned into 500 parties, namely, a total of 500 clients. During collaborative training of a global model, 100 clients are selected in each communication round.

CIFAR10 [15]: The Canadian Institute For Advanced Research (CIFAR10) dataset contains a small part of tiny images associated with 10 natural categories. It consists of 60,000 total samples, which are split into training data consisting of 50,000 samples and testing data consisting of 10,000 samples.

CIFAR100: The CIFAR100 dataset also contains a small part of tiny images like CIFAR10, whereas it has 100 categories, each of which consists of 600 samples. Besides, these 100 categories are divided into 20 super-categories. Thus, each sample in the CIFAR100 dataset has a “fine” label (the category it belongs to) and a “coarse” label (the super-category it belongs to).

EMNIST [9]: The EMNIST dataset is a set of handwritten character digits derived from the NIST Special Database

19 and converted to a 28×28 pixel image format and dataset structure that directly matches the MNIST dataset.

FMNIST [33]: The FMNIST dataset consists of Zalando’s article images that are related to 10 commodity categories. It is comprised of 70,000 total samples, which are split into training data consisting of 60,000 samples and testing data consisting of 10,000 samples.

2) *Attack Assumption*: RoseAgg is designed to mitigate targeted collusion attacks where the proportion of malicious clients and attack density are both high. To this end, the default setting establishes an attacker’s ratio of 0.5 in each adversarial round, and each communication round is designated as an adversarial round with a probability of 0.5. Additionally, multi-objective targeted model poisoning attacks can be orchestrated by the adversary $\mathcal{A}$ during adversarial rounds. For instance, $\mathcal{A}$ can inject multiple distinct backdoors to enhance attack success rates. In the process of locally constructing a poisoned dataset, each malicious client initiates by randomly sampling clean data from the training dataset. Subsequently, they proceed to inject targeted poisoned data samples into the preexisting clean data.

3) *Evaluation Metrics*: Two metrics are primarily considered for evaluating the effectiveness of targeted model poisoning attacks and defense methods: **ASR** - *attack success rate* indicates that the probability of the adversary $\mathcal{A}$ successfully inducing the global model to produce targeted misprediction. $\mathcal{A}$ aims to maximize ASR while an effective defense aims to prevent ASR from increasing. **TER** - *test error rate* indicates the error rate predicted by the global model on the test set. $\mathcal{A}$ ’s objective is to minimize the impact on the model’s test error rate in order to avoid easy detection. An effective defense should not affect the TER of the global model.

B. Effectiveness of Dynamic Clustering Component

Dynamic clustering is a crucial component of RoseAgg, involving the application of DBSCAN clustering to local updates. The subsequent stages of the pipeline are significantly influenced by the quality of the clustering result. Hence, the evaluation of this clustering step is prioritized in this section. In the evaluation of the dynamic clustering component, four metrics are considered for assessing its effectiveness. 1) #PC: the number of “poisoned clusters”, where a cluster is considered as “poisoned” if it contains poisoned updates. 2) #BC: the number of “benign clusters”, where a cluster is considered as “benign” if it does not include poisoned updates. 3) PCR: the proportion of “poisoning clusters” to all poisoning updates. A smaller PCR indicates a better clustering effect, as fewer “poisoned clusters” result in a smaller accumulation of credit scores in subsequent steps, leading to a diminished contribution of poisoned updates to the global model update. 4) BCR, the proportion of “benign clusters” to all benign updates. A larger BCR signifies a better clustering effect, as larger “benign clusters” lead to a greater accumulation of credit scores in subsequent steps, enlarging the contribution of benign updates to the global model update.

Table III demonstrates the results in a specific adversarial round. The dynamic clustering component consistently

$$\begin{aligned}
& + (G + 1 - \alpha) \|\nabla F(\mathbf{w}_{t-1}) - \nabla F(\mathbf{w}^*)\| + \Gamma \\
& \leq \sqrt{(1 + \alpha^2 L^2 - \alpha\mu) \|\mathbf{w}_{t-1} - \mathbf{w}^*\|} \\
& \quad + (G + 1)(8\Delta_2 \|\mathbf{w}_{t-1} - \mathbf{w}^*\| \\
& \quad + 4\Delta_1) + L(G + 1 - \alpha) \|\mathbf{w}_{t-1} - \mathbf{w}^*\| + \Gamma \\
& \leq (\sqrt{(1 + \alpha^2 L^2 - \alpha\mu) + 8\Delta_2(G + 1) + L(G + 1 - \alpha)}) \\
& \quad \|\mathbf{w}_{t-1} - \mathbf{w}^*\| + 4\Delta_1(G + 1) + \Gamma \quad (38)
\end{aligned}$$

递归地应用不等式 (38) 对于 T 全局迭代次数, 可以得出以下结论:

$$\|\mathbf{w}_T - \mathbf{w}^*\| \leq (1 - \rho)^T \|\mathbf{w}_0 - \mathbf{w}^*\| + (4\Delta_1(G + 1) + \Gamma)/\rho, \quad (39)$$

□

当 $|1 - \rho| < 1$ 时, 可以得出以下结论: $\lim_{T \rightarrow \infty} \|\mathbf{w}_T - \mathbf{w}^*\| \leq (4\Delta_1(G + 1) + \Gamma)/\rho$. 经过几次训练轮次后, 通过 RoseAgg 获得的全局模型与通过 FedAVG 获得的非攻击全局模型之间的差异将被限制在一个相对较窄的范围内。换句话说, RoseAgg 对联邦学习中的针对性共谋攻击具有鲁棒性。

VII. 实验结果

本节在四个基准数据集上对 RoseAgg 进行了全面评估, 旨在说明其有效性。结果证实了它在防御最先进的针对性模型中毒攻击方面的成功 [2], [3], [19], [31], [35], [39] (如第二节 B 所述) 在不同设置中的成功。同时, 在 RoseAgg 和最先进的防御措施 [6], [11], [22][23] (如第二节 C 所述) 之间进行了全面比较。

A. 实验设置

1) 数据集: 我们在评估中使用了四个基准图像分类数据集来说明 RoseAgg 的有效性。遵循先前使用参数为 0.5 的 Dirichlet 分布对训练数据进行划分的工作 [35]。具体来说, 训练数据被划分为 500 个部分, 即总共 500 个客户端。在全局模型的协同训练过程中, 每个通信轮次中会从 100 个客户端中选择。

CIFAR10 [15]: 加拿大高级研究院 (CIFAR10) 数据集包含一小部分与 10 个自然类别相关的微小图像。它包含 60,000 个总样本, 这些样本被分为包含 50,000 个样本的训练数据和包含 10,000 个样本的测试数据。

CIFAR100: CIFAR100 数据集也像 CIFAR10 一样包含一小部分微小图像, 但它有 100 个类别, 每个类别包含 600 个样本。此外, 这 100 个类别被分为 20 个超级类别。因此, CIFAR100 数据集集中的每个样本都有一个 “细粒度” 标签 (它所属的类别) 和一个 “粗粒度” 标签 (它所属的超级类别)。

EMNIST [9]: The EMNIST 数据集是一组手写数字字符, 源自 NIST 特殊数据库

19, 并转换为 28×28 像素图像格式和直接匹配 MNIST 数据集的数据集结构。

FMNIST [33]: The FMNIST 数据集由与 10 种商品类别相关的 Zalando 的文章图像组成。它包含总共 70,000 个样本, 这些样本分为 60,000 个样本的训练数据和 10,000 个样本的测试数据。

2) 攻击假设: RoseAgg 旨在缓解恶意客户端比例和攻击密度都较高的定向共谋攻击。为此, 默认设置在每个对抗轮次中建立攻击者的 0.5 比例, 并且每个通信轮次被指定为具有 0.5 概率的对抗轮次。此外, 在对抗轮次中, 攻击者 $\mathcal{A}$ 可以协调多目标定向模型中毒攻击。例如, $\mathcal{A}$ 可以注入多个不同的后门来提高攻击成功率。在本地构建中毒数据集的过程中, 每个恶意客户端首先从训练数据集中随机采样干净数据。随后, 他们继续将定向的中毒数据样本注入现有的干净数据中。

3) 评估指标: 主要考虑两个指标来评估针对性模型中毒攻击和防御方法的有效性: **ASR** - 攻击成功率 表示攻击者 $\mathcal{A}$ 成功诱导全局模型产生针对性错误预测的概率。 $\mathcal{A}$ 旨在最大化 ASR, 而有效的防御则旨在防止 ASR 增加。**TER** - 测试错误率 表示全局模型在测试集上的错误率。 $\mathcal{A}$ 的目标是尽量减少对模型测试错误率的影响, 以避免被轻易检测到。有效的防御不应影响全局模型的 TER。

B. 动态聚类组件的有效性

动态聚类是 RoseAgg 的一个关键组件, 涉及将 DBSCAN 聚类应用于本地更新。管道的后续阶段受到聚类结果质量的影响。因此, 本节优先评估这一聚类步骤。在动态聚类组件的评估中, 考虑了四个指标来评估其有效性。1) #PC: 中毒簇的数量, 其中簇被认为是 “中毒” 如果它包含中毒更新。2) #BC: 良性簇的数量, 其中簇被认为是 “良性” 如果它不包含中毒更新。3) PCR: 中毒簇占有中毒更新的比例。较小的 PCR 表示更好的聚类效果, 因为较少的 “中毒簇” 导致后续步骤中信用分数的累积较小, 从而减少了中毒更新对全局模型更新的贡献。4) BCR, 良性簇占有良性更新的比例。较大的 BCR 表示更好的聚类效果, 因为较大的 “良性簇” 导致后续步骤中信用分数的累积较大, 增加了良性更新对全局模型更新的贡献。

Table III demonstrates the results in a specific adversarial round. The dynamic clustering component consistently

TABLE II

EFFECTIVENESS OF ROSEAGG AND RECENT DEFENSES AGAINST STATE-OF-THE-ART ATTACKS FOR THE RESPECTIVE DATASET, IN TERMS OF ATTACK SUCCESS RATE (ASR) AND TEST ERROR RATE (TER). ALL METRIC VALUES ARE REPORTED AS PERCENTAGES

Attack	Dataset	Avg		RLR		FoolsGold		FLTrust		FLAME		RoseAgg	
		ASR	TER	ASR	TER	ASR	TER	ASR	TER	ASR	TER	ASR	TER
MR	CIFRA10	100	11.2	100	16.7	0	13.1	85.7	12.5	57.0	13.7	0	10.4
	EMNIST	100	1.9	98.4	1.5	0.78	1.7	2.34	1.7	0.98	2.1	0.59	1.69
	FMNIST	100	11.2	100	11.1	0	10.6	0	10.6	38.5	10.6	0	10.4
DBA	CIFRA10	100	10.9	82.4	12.4	100	11.6	27.3	12.5	3.37	13.4	3.32	10.4
	EMNIST	99.6	2.7	39	1.7	99.8	3.7	0.59	1.9	1.56	2.1	0.39	1.97
	FMNIST	100	11.4	100	11.3	100	13.7	0	10.5	0	10.5	0	10.3
FLIP	CIFRA10	100	20.8	100	20.5	2.59	12.2	64.5	21.4	97.9	22.4	3.13	10.4
	EMNIST	100	19.3	99.2	12.8	0	1.86	61.5	8.29	0	2.0	0	1.68
	FMNIST	98.8	20.1	98.2	20	1.95	10.6	3.52	10.6	97.9	19.9	1.95	10.5
EDGE	CIFRA10	97.5	12.7	99.0	24.1	1.53	13.0	54.6	13.1	5.61	13.8	3.57	10.5
	EMNIST	99	1.9	98	1.78	0	1.95	23	2.16	99	2.23	0	1.67
	FMNIST	99.0	11.2	99.0	11.2	2.05	10.5	6.00	10.7	99	12.0	1.00	10.5
NEUR	CIFRA10	96.5	13.1	97.4	15.4	2.04	11.6	46.9	12.8	1.02	13.5	1.57	10.6
	EMNIST	100	2.03	92	1.67	100	1.89	13	2.34	1.00	2.07	0	1.67
	FMNIST	99.0	11.0	99.0	11.3	41	10.6	2.25	10.7	97	10.7	2.00	10.4

TABLE III

EFFECTIVENESS OF DYNAMIC CLUSTERING COMPONENT. THE VALUES OF PCR AND BCR ARE IN PERCENTAGE

Attack	MR	DBA	FLIP	EDGE	NEUR	COMBINE
#PC	1	10	1	1	1	4
#BC	36	44	39	38	40	28
PCR	2	76.9	2	2	2	8
BCR	72	50.6	78	76	80	56

inclines towards consolidating poisoned updates into a limited number of clusters or a singular cluster. This phenomenon arises from the shared malicious goal among attackers, leading to a pronounced similarity in the feature spaces relevant to the malicious goal for each attacker. The measurement of this similarity is facilitated by calculating the pairwise cosine distance among local model updates.

C. Prevent Targeted Model Poisoning Attacks

A comparative analysis is conducted between RoseAgg and several existing defense mechanisms, namely RLR [23], FoolsGold [11], FLTurst [6], and FLAME [22], against state-of-the-art targeted model poisoning attacks including MR [2], DBA [35], FLIP [3], EDGE [31], and NEUR [39]. These evaluations are performed on three benchmark datasets. To assess the effectiveness of each defense approach, a global model is initially trained using the respective algorithm until it approaches convergence. Subsequently, an additional 60 rounds of experiments are conducted. During each round, the introduction of malicious clients is based on the aforementioned attack assumption. The largest attack success rate (ASR) in the 60 rounds and the corresponding test error rate (TER) are recorded and presented in Table II. Finally, a thorough analysis is then undertaken to assess the effectiveness of each defense method in mitigating the impact of targeted model poisoning attacks.

Table II demonstrates the robustness of RoseAgg against all types of targeted model poisoning attacks in the respective dataset. Furthermore, RoseAgg ensures that the performance of the global model is not compromised. This is achieved by

considering the contribution of all local model updates during the model aggregation process on the server.

However, RLR is not effective in preventing most types of attacks. This is due to the assumption that the number of malicious clients remains significantly below a threshold value θ . Once the number of malicious clients surpasses this threshold, the model deviates from the benign direction and attempts to move toward the malicious direction, resulting in a failure of defense.

While FoolsGold can prevent most types of attacks, it fails to defend against DBA. Since DBA spreads the attacks across different rounds, FoolsGold struggles to distinguish between poisoned and benign model updates effectively. Consequently, DBA can successfully evade the detection and defense mechanisms of FoolsGold, leading to specific mispredictions.

Furthermore, FLTrust lacks robustness on the CIFAR10 dataset. The substantial divergence between the distribution of the root set on the server and the overall data distribution hinders FLTrust's ability to calculate and maintain a representative model update. As a result, FLTrust may mistakenly remove benign local updates while retaining malicious ones.

In the case of FLAME, when the number of malicious clients is high, a significant portion of poisoned model updates may be erroneously clustered together with benign model updates. Consequently, this clustering process may drive the model towards a malicious direction during the aggregation phase. Furthermore, FLAME introduces noise into the model, thereby deteriorating the accuracy of the global model.

D. Prevent Multi-Objective Attacks

To enhance the success rate of the attack, the adversary $\mathcal{A}$ can employ a multi-objective targeted model poisoning attack. For instance, $\mathcal{A}$ can inject multiple backdoors simultaneously by utilizing distinct client groups [22]. The poisoned model updates generated in an adversarial round do not display typical patterns and are challenging to differentiate from benign model updates. Consequently, preventing multi-objective attacks becomes a formidable task.

We conduct an evaluation to assess the effectiveness of RoseAgg and other defense mechanisms against a

表 II ROSEAGG 和最近防御措施在各自数据集上针对最先进攻击的有效性，以攻击成功率 (ASR) 和测试错误率 (TER) 为指标。所有指标值均以百分比报告

Attack	Dataset	Avg		RLR		FoolsGold		FLTrust		FLAME		RoseAgg	
		ASR	TER	ASR	TER	ASR	TER	ASR	TER	ASR	TER	ASR	TER
MR	CIFRA10	100	11.2	100	16.7	0	13.1	85.7	12.5	57.0	13.7	0	10.4
	EMNIST	100	1.9	98.4	1.5	0.78	1.7	2.34	1.7	0.98	2.1	0.59	1.69
	FMNIST	100	11.2	100	11.1	0	10.6	0	10.6	38.5	10.6	0	10.4
DBA	CIFRA10	100	10.9	82.4	12.4	100	11.6	27.3	12.5	3.37	13.4	3.32	10.4
	EMNIST	99.6	2.7	39	1.7	99.8	3.7	0.59	1.9	1.56	2.1	0.39	1.97
	FMNIST	100	11.4	100	11.3	100	13.7	0	10.5	0	10.5	0	10.3
FLIP	CIFRA10	100	20.8	100	20.5	2.59	12.2	64.5	21.4	97.9	22.4	3.13	10.4
	EMNIST	100	19.3	99.2	12.8	0	1.86	61.5	8.29	0	2.0	0	1.68
	FMNIST	98.8	20.1	98.2	20	1.95	10.6	3.52	10.6	97.9	19.9	1.95	10.5
EDGE	CIFRA10	97.5	12.7	99.0	24.1	1.53	13.0	54.6	13.1	5.61	13.8	3.57	10.5
	EMNIST	99	1.9	98	1.78	0	1.95	23	2.16	99	2.23	0	1.67
	FMNIST	99.0	11.2	99.0	11.2	2.05	10.5	6.00	10.7	99	12.0	1.00	10.5
NEUR	CIFRA10	96.5	13.1	97.4	15.4	2.04	11.6	46.9	12.8	1.02	13.5	1.57	10.6
	EMNIST	100	2.03	92	1.67	100	1.89	13	2.34	1.00	2.07	0	1.67
	FMNIST	99.0	11.0	99.0	11.3	41	10.6	2.25	10.7	97	10.7	2.00	10.4

表 III 动态聚类组件的有效性。PCR 和 BCR 的值以百分比表示

Attack	MR	DBA	FLIP	EDGE	NEUR	COMBINE
#PC	1	10	1	1	1	4
#BC	36	44	39	38	40	28
PCR	2	76.9	2	2	2	8
BCR	72	50.6	78	76	80	56

倾向于将中毒更新合并到有限数量的簇或单个簇中。这种现象源于攻击者之间的共同恶意目标，导致每个攻击者与恶意目标相关的特征空间中存在显著相似性。这种相似性的测量是通过计算局部模型更新之间的成对余弦距离来实现的。

C. 防止针对模型的中毒攻击

对 RoseAgg 和几种现有的防御机制，即 RLR [23], FoolsGold [11], FLTurst [6], 和 FLAME [22], 进行了比较分析，以抵抗最先进的针对模型中毒攻击，包括 MR [2], DBA [35], FLIP [3], EDGE [31], 和 NEUR [39]。这些评估是在三个基准数据集上进行的。为了评估每种防御方法的有效性，首先使用相应的算法训练全局模型，直到接近收敛。随后，进行额外的 60 轮实验。在每一轮中，恶意客户端的引入基于上述攻击假设。记录并展示在 60 轮中最大的攻击成功率（ASR）和相应的测试错误率（TER），如表 II 所示。最后，对每种防御方法的有效性进行全面分析，以评估其在减轻针对模型中毒攻击影响方面的效果。

表 II 展示了 RoseAgg 在各个数据集上对所有类型的针对模型中毒攻击的鲁棒性。此外，RoseAgg 确保全局模型的性能不受影响。这是通过

考虑到在服务器上的模型聚合过程中，所有本地模型更新的贡献。

然而，RLR 在防止大多数类型的攻击方面并不有效。这是由于假设恶意客户端的数量显著低于阈值 θ 。一旦恶意客户端的数量超过这个阈值，模型就会偏离良性方向并试图向恶意方向移动，从而导致防御失败。

虽然 FoolsGold 可以防止大多数类型的攻击，但它无法防御 DBA。由于 DBA 将攻击分散到不同的轮次中，FoolsGold 难以有效地区分中毒和良性模型更新。因此，DBA 可以成功规避 FoolsGold 的检测和防御机制，导致特定的错误预测。

此外，FLTrust 在 CIFAR10 数据集上缺乏鲁棒性。服务器上根集合的分布与整体数据分布之间的巨大差异阻碍了 FLTrust 计算和维护代表性模型更新的能力。因此，FLTrust 可能会错误地移除良性本地更新，同时保留恶意更新。

在 FLAME 的情况下，当恶意客户端的数量很高时，很大一部分被污染的模型更新可能会被错误地与良性模型更新聚类在一起。因此，这种聚类过程可能会在聚合阶段将模型推向恶意方向。此外，FLAME 将噪声引入模型，从而降低了全局模型的准确性。

D. 防止多目标攻击

为了提高攻击的成功率，攻击者 $\mathcal{A}$ 可以采用多目标定向模型污染攻击。例如， $\mathcal{A}$ 可以通过利用不同的客户端组 [22] 同时注入多个后门。在对抗性轮次中生成的被污染的模型更新不具有典型模式，并且难以与良性模型更新区分开来。因此，防止多目标攻击成为一项艰巨的任务。

我们进行评估，以评估 RoseAgg 和其他防御机制对

TABLE IV
EFFECTIVENESS OF ROSEAGG AND RECENT DEFENSES AGAINST MULTI-OBJECTIVE TARGETED MODEL POISONING ATTACKS FOR THE RESPECTIVE DATASET, IN TERMS OF ATTACK SUCCESS RATE (ASR) AND TEST ERROR RATE (TER). ALL METRIC VALUES ARE REPORTED AS PERCENTAGES

Dataset	Objective	Avg		RLR		FoolsGold		FLTrust		FLAME		RoseAgg	
		ASR	TER	ASR	TER	ASR	TER	ASR	TER	ASR	TER	ASR	TER
CIFAR10	O1	100		100		52.7		3.32		1.95		2.15	
	O2	98.6	10.6	49	14.6	10.5		1.17		0.98		0.39	10.5
	O3	99.4		68		12.4	11.6	2.73	12.4	0.39	13.6	0.39	
	O4	94.9		31		1.98		0		0.39		0	
EMNIST	O1	100		54.9		100		0.78		0.19		0.59	
	O2	99.0	1.30	0.20	1.60	98.0		0		0		0	1.97
	O3	99.8		0		99.2	1.28	4.1	2.91	0.39	2.05	0	
	O4’	97.3		0		96.5		0.59		1.17		0.59	
FMNIST	O1	100		100		100		0.97		1.17		1.17	
	O2	79.9	11.3	69.5	11.2	71.8	10.6	0.20	10.8	0.20	10.5	0.20	10.4
	O3	68.7		66.2		76.1		2.35		1.96		1.76	
	O4’	86.2		78.9		92.1		0.39		0.40		0.39	

multi-objective attack on the CIFAR10 dataset. The defense effect is presented in Table IV. This evaluation encompasses four malicious objectives within a single round of attack. The objectives, denoted as O1, O2, O3, and O4’, indicate that $\mathcal{A}$ intends to misclassify examples containing a pixel-pattern trigger “F”, “L”, “A”, and “I” as classes “0”, “1”, “6”, and “2”, respectively. Objective O4 represents the attacker’s goal of misclassifying images of cars with vertically striped walls as birds.

Table IV illustrates the robustness of all defenses against multi-objective targeted model poisoning attacks. Due to the distinct targets of each malicious client, the poisoned model update gradients exhibit a weak correlation in direction. Although the adaptive partial aggregating module cannot effectively mitigate the impact of potentially malicious model updates from a directional perspective, the clean ingredient extraction module compensates for this limitation. Given that the number of “poisoned clusters” is smaller than that of “benign clusters”, the clean ingredient extraction module ensures that the summation of projection variances of benign local model updates surpasses that of poisoned local model updates. This implies that the influence of benign model updates outweighs that of malicious model updates, thereby guiding the model toward the benign direction and mitigating the impact of targeted model poisoning attacks.

FoolsGold fails to effectively prevent multi-objective targeted model poisoning attacks. The reason is that poisoned model updates within a single round lack common characteristics, making it challenging for FoolsGold to distinguish between poisoned and benign model updates. Consequently, the defense mechanism of FoolsGold is rendered ineffective. RLR is not robust to multi-objective targeted model poisoning attacks, as the number of malicious clients surpasses the threshold allowed by RLR, resulting in defense failure. Additionally, while FLTrust and FLAME successfully mitigate multi-objective attacks, they do experience varying degrees of accuracy loss under specific scenarios, such as on the CIFAR10 dataset.

E. Defense at Different Convergence Stages

Malicious clients can be selected at any stage of federated learning. Therefore, studying targeted model poisoning attacks

and defenses at different stages provides insights into the generalizability of defense mechanisms. Defense techniques that perform well at one stage of convergence may exhibit different efficacy at other stages. Hence, it is crucial to assess the efficacy of defenses at various convergence stages of the global model. Figure 5(a) demonstrates the defensive impact of RoseAgg and other defenses in countering the “EDGE” attack at different rounds on the CIFAR10 dataset.

RoseAgg and FoolsGold demonstrate robustness throughout the convergence stages of the global model. In contrast, RLR and FLTrust fail to effectively prevent targeted model poisoning attacks during the entire process of federated training. Furthermore, FLAME fails to prevent targeted model piosoning attacks at the early and intermediate stages of convergence. This observation highlights the adversary’s ability to evade defense measures different stages. Hence, evaluating the effectiveness of defense approaches during different convergence stages becomes essential when developing countermeasures against targeted model poisoning attacks.

F. Impact of the Degree of Non-IID Data

The adaptive partial aggregating module of RoseAgg incorporates the DBSCAN algorithm, which clusters local updates based on cosine distances between pairwise updates. However, the distribution of data among clients can influence the cluster results, potentially impacting the effectiveness of RoseAgg. To assess of the effectiveness of RoseAgg under varying degrees of non-IID data among clients, three experiments are performed on the CIFAR10 dataset. These experiments demonstrate the effectiveness of RoseAgg against “MR”, “DBA”, and “EDGE” attacks. Following previous works [2], [35], the degree of non-IID data across clients is manipulated by adjusting the α parameter of the Dirichlet distribution.

Figure 5(c) presents the evaluation results of RoseAgg in defending against “MR”, “DBA”, and “EDGE” attacks. The results demonstrate that RoseAgg effectively safeguards against targeted model poisoning attacks under varying degrees of non-IID data among clients. The success of RoseAgg can be attributed to its ability to identify and address common characteristics shared by malicious clients. These shared characteristics often involve similarities in direction or magnification of the magnitude. Fortunately, the adaptive

表 IV RoseAgg 和最近防御措施在多目标针对性模型中毒攻击方面的有效性，针对各自数据集，以攻击成功率（ASR）和测试错误率（TER）衡量。所有指标值均以百分比报告

Dataset	Objective	Avg		RLR		FoolsGold		FLTrust		FLAME		RoseAgg	
		ASR	TER	ASR	TER	ASR	TER	ASR	TER	ASR	TER	ASR	TER
CIFAR10	O1	100		100		52.7		3.32		1.95		2.15	
	O2	98.6	10.6	49	14.6	10.5		1.17		0.98		0.39	10.5
	O3	99.4		68		12.4	11.6	2.73	12.4	0.39	13.6	0.39	
	O4	94.9		31		1.98		0		0.39		0	
EMNIST	O1	100		54.9		100		0.78		0.19		0.59	
	O2	99.0	1.30	0.20	1.60	98.0		0		0		0	1.97
	O3	99.8		0		99.2	1.28	4.1	2.91	0.39	2.05	0	
	O4’	97.3		0		96.5		0.59		1.17		0.59	
FMNIST	O1	100		100		100		0.97		1.17		1.17	
	O2	79.9	11.3	69.5	11.2	71.8	10.6	0.20	10.8	0.20	10.5	0.20	10.4
	O3	68.7		66.2		76.1		2.35		1.96		1.76	
	O4’	86.2		78.9		92.1		0.39		0.40		0.39	

在 CIFAR10 数据集上的多目标攻击。防御效果展示在表 IV。此评估包含单个攻击轮次内的四个恶意目标。这些目标表示为 O1、O2、O3 和 O4’，表明 $\mathcal{A}$ 打算将包含像素模式触发器 “F”、“L”、“A” 和 “I” 的示例误分类为类别 “0”、“1”、“6” 和 “2”，分别。目标 O4 代表攻击者将带有垂直条纹墙壁的汽车图像误分类为鸟类的目标。

TableIV 展示了所有防御措施对多目标定向模型中毒攻击的鲁棒性。由于每个恶意客户端的目标不同，中毒的模型更新梯度在方向上表现出弱相关性。尽管自适应部分聚合模块不能从方向上有效减轻潜在恶意模型更新的影响，但干净成分提取模块弥补了这一局限性。考虑到 “中毒簇” 的数量小于 “良性簇” 的数量，干净成分提取模块确保良性本地模型更新的投影方差之和超过中毒本地模型更新的投影方差之和。这意味着良性模型更新的影响大于恶意模型更新的影响，从而引导模型朝良性方向发展，减轻定向模型中毒攻击的影响。

FoolsGold 无法有效防止多目标定向模型中毒攻击。原因是单轮次内的中毒模型更新缺乏共同特征，使得 FoolsGold 难以区分中毒和良性模型更新。因此，FoolsGold 的防御机制失效。RLR 对多目标定向模型中毒攻击不鲁棒，因为恶意客户端的数量超过了 RLR 允许的阈值，导致防御失败。此外，虽然 FLTrust 和 FLAME 成功减轻了多目标攻击，但在特定场景下，如 CIFAR10 数据集上，它们仍会不同程度地损失精度。

E. 在不同收敛阶段进行防御

恶意客户端可以在联邦学习的任何阶段被选中。因此，研究针对模型中毒攻击

以及在各个阶段的防御，可以提供对防御机制泛化能力的见解。在收敛的一个阶段表现良好的防御技术，在其他阶段可能表现出不同的有效性。因此，评估防御机制在全球模型不同收敛阶段的功效至关重要。图 5(a)demonstrates the defensive impact of RoseAgg and other defenses in countering the “EDGE” attack at different rounds on the CIFAR10 dataset.

RoseAgg 和 FoolsGold 在全球模型的收敛阶段表现出鲁棒性。相比之下，RLR 和 FLTrust 在整个联邦训练过程中无法有效防止针对模型的中毒攻击。此外，FLAME 在收敛的早期和中期阶段无法防止针对模型的中毒攻击。这一观察结果表明攻击者能够规避不同阶段的防御措施。因此，在开发针对模型中毒攻击的对抗措施时，评估防御方法在不同收敛阶段的有效性变得至关重要。

F. 非独立同分布数据程度的影响

RoseAgg 的自适应部分聚合模块集成了 DBSCAN 算法，该算法根据成对更新之间的余弦距离对本地更新进行聚类。然而，客户端之间的数据分布会影响聚类结果，从而可能影响 RoseAgg 的有效性。为了评估 RoseAgg 在客户端之间非独立同分布数据程度变化下的有效性，在 CIFAR10 数据集上进行了三项实验。这些实验展示了 RoseAgg 对 “MR”、“DBA” 和 “EDGE” 攻击的有效性。遵循先前工作 [2][35]，通过调整 Dirichlet 分布的 α 参数来操纵客户端之间的非独立同分布数据程度。

Figure 5(c) presents the evaluation results of RoseAgg in defending against “MR”, “DBA”, and “EDGE” attacks. The results demonstrate that RoseAgg effectively safeguards against targeted model poisoning attacks under varying degrees of non-IID data among clients. The success of RoseAgg can be attributed to its ability to identify and address common characteristics shared by malicious clients. These shared characteristics often involve similarities in direction or magnification of the magnitude. Fortunately, the adaptive

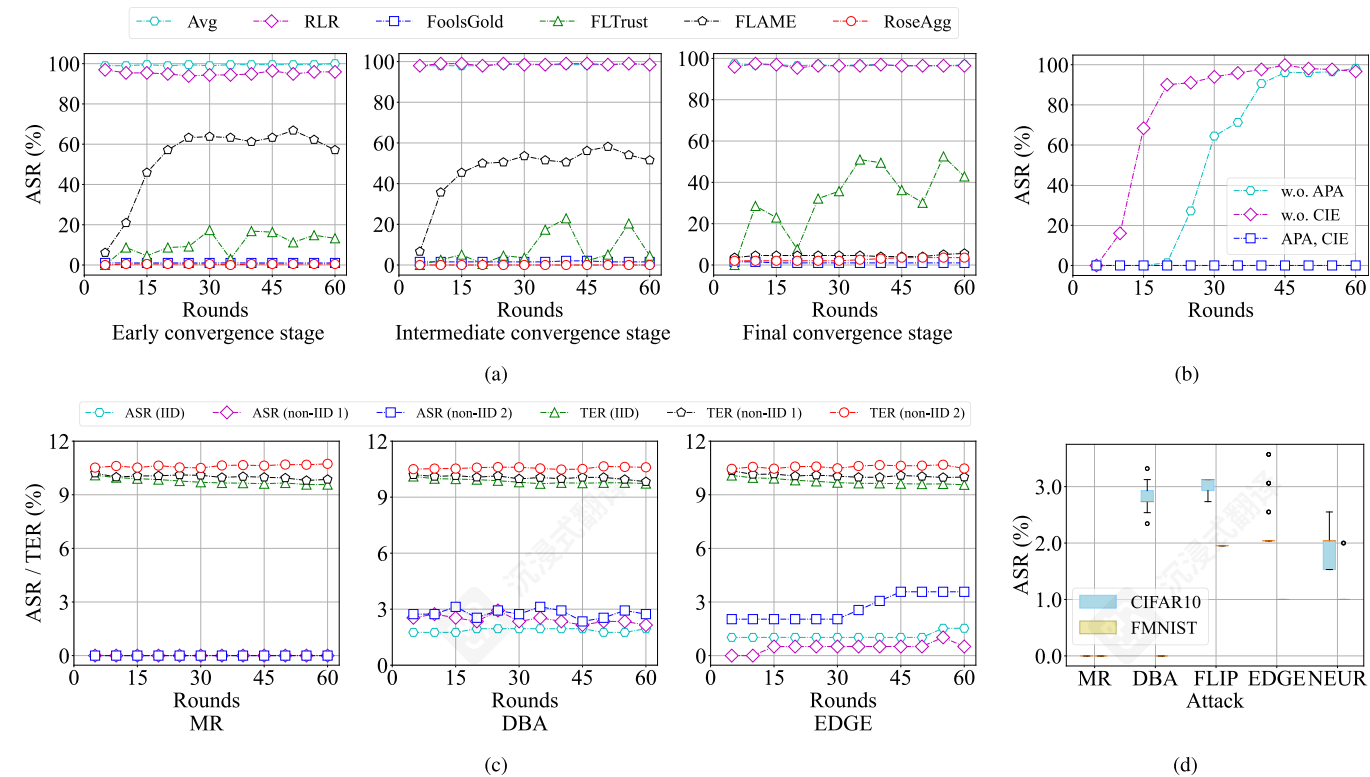


Fig. 5. Defense against state-of-the-art targeted model poisoning attacks. (a) Attack success rate of “EDGE” attack at different convergence stages. (b) Effectiveness of RoseAgg’s components. Specifically, the “w.o. APA” and “w.o. CIE” configurations involve disabling the adaptive partial aggregating and clean ingredient extraction modules. (c) Attack success rate and test error rate of RoseAgg in defending against attacks under varying degrees of non-IID data among clients. The hyperparameter α of the Dirichlet distribution adapted by non-IID 1 and non-IID 2 is 0.9 and 0.5, respectively. (d) Stability of RoseAgg.

partial aggregating module within RoseAgg mitigates these common characteristics, thereby neutralizing their impact. Furthermore, the calculation of the clean ingredient ensures that the cumulative weight of benign local model updates surpasses that of malicious model updates. Consequently, the model becomes biased in a favorable direction, effectively preventing targeted model poisoning attacks.

G. Stability of RoseAgg

The stability of RoseAgg across repeated randomized experiments holds paramount significance in establishing the reliability and robustness of its performance. Through multiple iterations of randomized trials, RoseAgg consistently demonstrates its effectiveness in defending against targeted model poisoning attacks. This stability implies that the algorithm’s success is not contingent on specific, isolated conditions but persists across various scenarios. Specifically, the utilization of boxplots in Figure 5(d) provides a clear visualization of the stability of RoseAgg. Various attacks are applied across the CIFAR10 and FMNIST dataset, with the experiment replicated 30 times. The observed discrepancy between the maximum and minimum ASR remains within a narrow margin, *i.e.*, not surpassing 1.57%.

H. Effectiveness of RoseAgg’s Components

We analyze and evaluate each individual component of RoseAgg to assess its contribution toward its defense.

Figure 5(b) provides an intuitive evaluation result for each component. Firstly, we disable the adaptive partial aggregating module and compare the effectiveness of defense with the full implementation of RoseAgg. In this scenario, we observe that RoseAgg fails to effectively defend against “MR” attacks. Without the adaptive partial aggregating module, RoseAgg is unable to identify and mitigate the impact of local model updates with amplified magnitudes or similar directional contributions. Unfortunately, these characteristics are commonly found in most poisoned model updates. Secondly, we disable the clean ingredient extraction module and instead directly calculate an average of the model updates handled by the adaptive partial aggregating module. Similarly, RoseAgg fails to provide adequate defense against “MR” attacks. The simplistic averaging algorithm fails to ensure that the cumulative weight of benign local model updates surpasses that of the poisoned model updates, ultimately leading to defense failure.

I. Impact of Number of Malicious Clients

Furthermore, in alignment with the core motivations of this paper, additional experiments have been incorporated to investigate the impact of varying the number of malicious clients. This deliberate variation serves to delve deeper into the nuanced dynamics and robustness of RoseAgg under different scenarios. Specifically, Figure 6 illustrates experiments conducted on the CIFAR-100 dataset, evaluating RoseAgg’s defense against CerP by varying the percentage of malicious

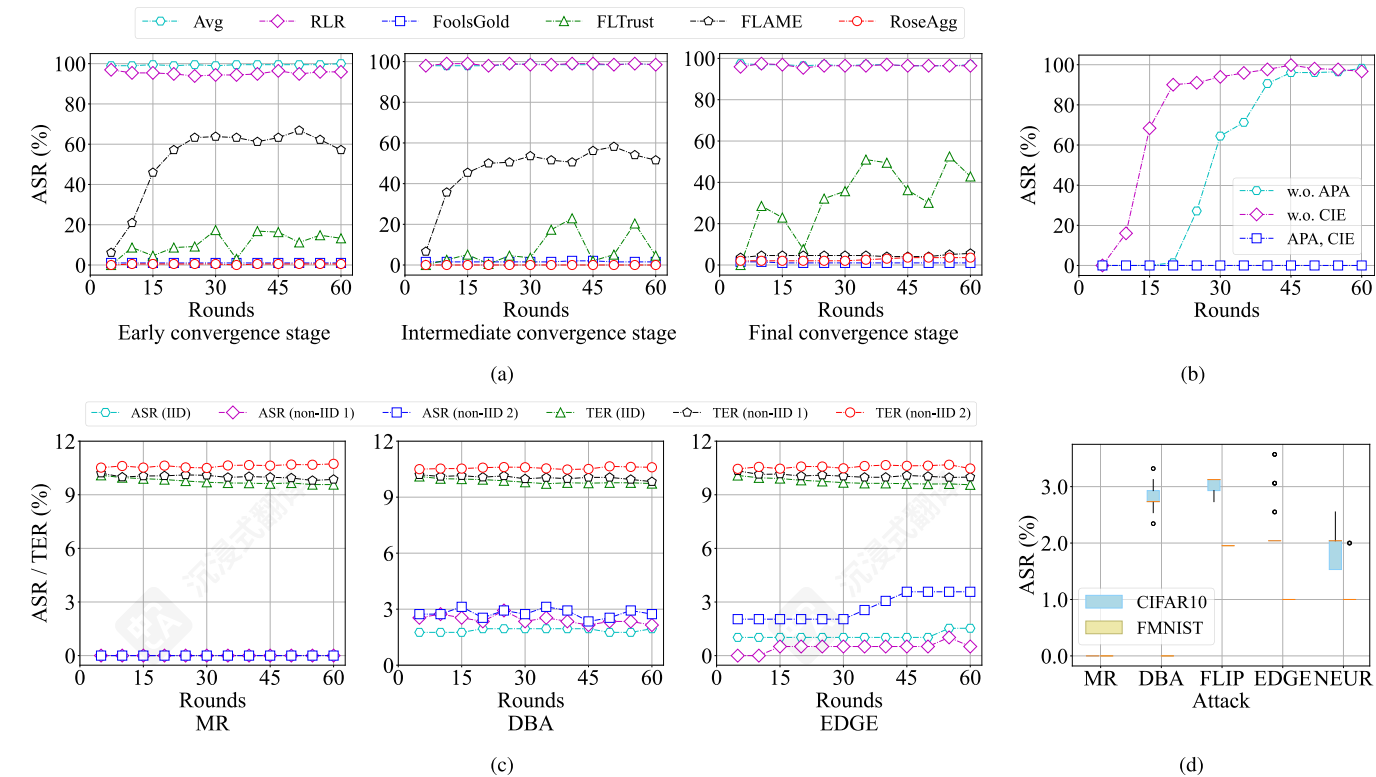


图 5. 针对最先进的目标模型中毒攻击的防御。(a) “EDGE” 攻击在不同收敛阶段的攻击成功率。(b) RoseAgg 组件的有效性。具体来说, “w.o. APA” 和 “w.o. CIE” 配置涉及禁用自适应部分聚合和清洁成分提取模块。(c) RoseAgg 在防御不同攻击时, 在客户端之间不同程度非独立同分布 (non-IID) 数据下的攻击成功率和测试错误率。非独立同分布 1 和非独立同分布 2 的 Dirichlet 分布的超参数 α 分别为 0.9 和 0.5。(d) RoseAgg 的稳定性。

RoseAgg 中的部分聚合模块缓解了这些常见特征, 从而消除了它们的影响。此外, 清洁成分的计算确保了良性本地模型更新的累积权重超过恶意模型更新的累积权重。因此, 模型在有利方向上产生偏差, 有效防止了目标模型中毒攻击。

G. RoseAgg 的稳定性

RoseAgg 在重复随机实验中的稳定性对于建立其性能的可靠性和鲁棒性具有至关重要的意义。通过多次随机试验的迭代, RoseAgg 始终如一地展示出其在防御针对模型中毒攻击方面的有效性。这种稳定性表明, 算法的成功并非依赖于特定的、孤立的条件, 而是在各种场景中持续存在。具体而言, 图 5(d) 中的箱线图清晰地展示了 RoseAgg 的稳定性。在各种攻击应用于 CIFAR10 和 FMNIST 数据集的情况下, 实验重复进行了 30 次。观察到的最大值和最小值 ASR 之间的差异保持在狭窄的范围内, 即, 不超过 1.57%。

H. RoseAgg 组件的有效性

我们分析和评估 RoseAgg 的每个独立组件, 以评估其对防御的贡献。

图 5(b) 为每个组件提供了一个直观的评估结果。首先, 我们禁用自适应部分聚合模块, 并将防御效果与 RoseAgg 的完整实现进行比较。在这种情况下, 我们观察到 RoseAgg 无法有效防御 “MR” 攻击。没有自适应部分聚合模块, RoseAgg 无法识别和减轻具有放大幅度或相似方向贡献的本地模型更新的影响。不幸的是, 这些特征在大多数中毒模型更新中很常见。其次, 我们禁用干净成分提取模块, 并直接计算由自适应部分聚合模块处理的模型更新的平均值。同样, RoseAgg 无法提供足够的防御 “MR” 攻击。简单的平均算法无法确保良性本地模型更新的累积权重超过中毒模型更新的权重, 最终导致防御失败。

I. 恶意客户端数量的影响

此外, 为了契合本文的核心动机, 我们补充了额外的实验来研究恶意客户端数量变化的影响。这种有意的变化有助于更深入地探究 RoseAgg 在不同场景下的复杂动态和鲁棒性。具体而言, 图 6 展示了在 CIFAR-100 数据集上进行的实验, 评估了 RoseAgg 对 CerP 的防御能力, 通过改变恶意客户端的百分比

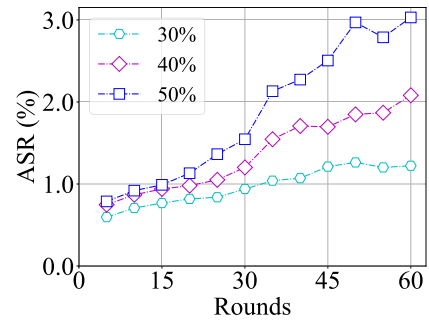


Fig. 6. Impact of the number of malicious clients. “30%”, “40%”, and “50%” represent the proportion of malicious clients is 30%, 40%, and 50%.

clients. As the proportion of malicious clients increases, an upward trend in the ASR is observed, yet it consistently remains within a relatively narrow range, hovering around three percent.

J. Computational Complexity Analysis

The computational complexity of RoseAgg, particularly in comparison to the widely used Federated Averaging (FedAVG), is a critical aspect of evaluating its practical viability. RoseAgg mainly introduces additional steps in the form of DBSCAN clustering and a PCA process to enhance its defense against targeted model poisoning attacks. Therefore, the computational complexity analysis of RoseAgg involves assessing the additional time complexity introduced by the clustering using DBSCAN and the PCA process.

In the DBSCAN clustering, the calculation of the distance matrix and neighbor query dominates the time complexity, resulting in $O(M^2d)$, where d is the dimensionality of the local updates. Additionally, the PCA process introduces time complexity mainly through Singular Value Decomposition (SVD), which is not more than $O(d^2M)$. In the context of general federated learning, the dimension of deep neural networks is significantly larger than the number of participants. Consequently, the extra time complexity of RoseAgg compared to FedAVG is approximately $O(d^2M)$. It’s important to note that both DBSCAN clustering and PCA operate separately according to the neural network hierarchy. Consequently, the combined time complexity is reduced to $O(P^2M)$, where P is the number of parameters in the layer with the most parameters. In the context of federated processes, the additional time complexity introduced by RoseAgg is deemed negligible compared to the computational demands of training deep neural networks. Therefore, RoseAgg remains computationally feasible for practical applications.

VIII. CONCLUSION

In this paper, RoseAgg is introduced as a defense mechanism against targeted model poisoning attacks, ensuring effective defense without compromising the global model’s performance. An adaptive partial aggregating strategy is devised to identify and restrict common characteristics found in poisoned model updates. Furthermore, a dimension-reduction method is employed to analyze a plausible clean ingredient, facilitating the bootstrapping of credit scores for

each local update. Finally, the consideration of both potentially malicious and benign local updates ensures the preservation of global model performance. The effectiveness of RoseAgg is theoretically illustrated, and extensive evaluations demonstrate its robustness against seven advanced attacks, even under conditions of a relatively high proportion of malicious clients and attack density.

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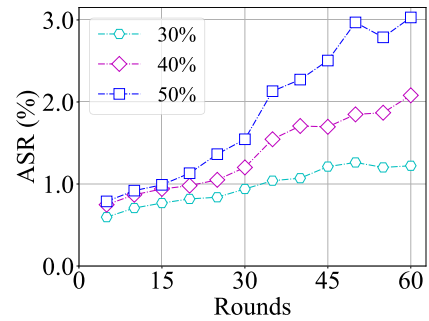


图 6. 恶意客户端数量的影响。 “30%”， “40%” 和 “50%” 表示恶意客户端的比例分别为 30%， 40% 和 50%。

客户端。随着恶意客户端比例的增加，观察到 ASR 呈上升趋势，但始终保持在相对较窄的范围内，徘徊在三个百分点左右。

J. 计算复杂度分析

RoseAgg 的计算复杂度，尤其是在与广泛使用的联邦平均（FedAVG）相比时，是其实际可行性的关键方面。RoseAgg 主要通过 DBSCAN 聚类 and PCA 过程引入额外的步骤，以增强其针对模型中毒攻击的防御。因此，RoseAgg 的计算复杂度分析涉及评估 DBSCAN 聚类和 PCA 过程引入的额外时间复杂度。

在 DBSCAN 聚类中，距离矩阵的计算和邻居查询主导了时间复杂度，导致 $O(M^2d)$ ，其中 d 是局部更新的维度。此外，PCA 过程主要通过奇异值分解（SVD）引入时间复杂度，不超过 $O(d^2M)$ 。在通用联邦学习的背景下，深度神经网络的维度远大于参与者数量。因此，与 FedAVG 相比，RoseAgg 的额外时间复杂度约为 $O(d^2M)$ 。需要注意的是，DBSCAN 聚类和 PCA 根据神经网络层次结构分别操作。因此，组合时间复杂度降低到 $O(P^2M)$ ，其中 P 是参数最多的层中的参数数量。在联邦处理背景下，RoseAgg 引入的额外时间复杂度与训练深度神经网络的计算需求相比可忽略不计。因此，RoseAgg 在实际应用中仍然具有计算可行性。

第八章 . 结论

在本文中，RoseAgg 被介绍为一种针对目标模型中毒攻击的防御机制，确保有效防御而不影响全局模型的性能。设计了一种自适应的部分聚合策略来识别和限制中毒模型更新中常见的特征。此外，采用了一种降维方法来分析可能的干净成分，从而为每个本地更新构建信用评分。

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